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1. CBSE X 2021

A. Formulae

1.1.1 (Line) The normal form of the equation of a line is

$$\mathbf{n}^\top \mathbf{x} = c \tag{1.1.1.1}$$

and its parametric form is

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}, \quad \kappa \in \mathbb{R}, \tag{1.1.1.2}$$

where

$$\mathbf{m}^\top \mathbf{n} = 0, \quad \mathbf{n}^\top \mathbf{h} = c \tag{1.1.1.3}$$

A point $\mathbf{x}_0$ lies on the line iff $\mathbf{n}^\top \mathbf{x}_0 = c$. The intersection with the y-axis is obtained by setting $\mathbf{e}_1^\top \mathbf{x} = 0$.

1.1.2 (Circle) A circle with centre $\mathbf{c}$ and radius r is the conic

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{1.1.2.1}$$

with

$$\mathbf{V} = \mathbf{I}, \quad \mathbf{u} = -\mathbf{c}, \quad f = \|\mathbf{c}\|^2 - r^2 \tag{1.1.2.2}$$

Solution: A point $\mathbf{x}$ is at distance r from $\mathbf{c}$ iff

$$\|\mathbf{x} - \mathbf{c}\|^2 = \mathbf{x}^\top \mathbf{x} - 2\mathbf{c}^\top \mathbf{x} + \|\mathbf{c}\|^2 - r^2 = 0 \tag{1.1.2.3}$$

which yields (1.1.2.2) on comparing with (1.1.2.1).

1.1.3 (Line-Conic Intersection, Chord and Tangent) Let $g(\mathbf{x}) = \mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f$. The line (1.1.1.2) meets the conic (1.1.2.1) at $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$, where

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) \pm \sqrt{\Delta}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}}, \quad \Delta = \left[\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u})\right]^2 - g(\mathbf{h}) \left(\mathbf{m}^\top \mathbf{V} \mathbf{m}\right) \tag{1.1.3.1}$$

The length of the chord and the tangency condition are

$$\|\mathbf{x}_1 - \mathbf{x}_2\| = |\kappa_1 - \kappa_2| \, \|\mathbf{m}\| = \frac{2\sqrt{\Delta}}{|\mathbf{m}^\top \mathbf{V} \mathbf{m}|} \|\mathbf{m}\|, \qquad \text{tangent} \iff \Delta = 0 \tag{1.1.3.2}$$

Solution: Substituting $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ in $g(\mathbf{x}) = 0$,

$$\left(\mathbf{m}^\top \mathbf{V} \mathbf{m}\right) \kappa^2 + 2\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) \kappa + g(\mathbf{h}) = 0 \tag{1.1.3.3}$$

whose roots are (1.1.3.1). The two points coincide (the line touches the conic) iff the discriminant vanishes, and $\mathbf{x}_1 - \mathbf{x}_2 = (\kappa_1 - \kappa_2)\, \mathbf{m}$ gives (1.1.3.2).

1.1.4 (Angle between vectors, Tangent and Radius) The angle θ between $\mathbf{a}$ and $\mathbf{b}$ is

$$\cos\theta = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|} \tag{1.1.4.1}$$

If a line $\mathbf{x} = \mathbf{P} + \kappa\mathbf{m}$ is tangent to a circle with centre $\mathbf{O}$ at the point $\mathbf{P}$ on the circle, then

$$\mathbf{m}^\top (\mathbf{P} - \mathbf{O}) = 0 \tag{1.1.4.2}$$

Solution: From (1.1.3.2) with $\mathbf{V} = \mathbf{I}$, $\mathbf{u} = -\mathbf{O}$, $\mathbf{h} = \mathbf{P}$ and $g(\mathbf{P}) = 0$ (since $\mathbf{P}$ is on the circle), $\Delta = \left[\mathbf{m}^\top (\mathbf{P} - \mathbf{O})\right]^2 = 0$, yielding (1.1.4.2).

1.1.5 (System of Linear Equations) The pair of lines $a_1x+b_1y+c_1 = 0$, $a_2x+b_2y+c_2 = 0$ is the matrix equation

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{A} = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -c_1 \\ -c_2 \end{pmatrix} \tag{1.1.5.1}$$

It is consistent iff $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}([\mathbf{A} \mid \mathbf{b}])$. Hence (taking $a_2, b_2, c_2 \neq 0$ in the ratios)

$$\left|\mathbf{A}\right| \neq 0 \iff \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \implies \text{intersecting} \tag{1.1.5.2}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \implies \operatorname{rank}(\mathbf{A}) = \operatorname{rank}([\mathbf{A} \mid \mathbf{b}]) = 1 \text{ (coincident)} \tag{1.1.5.3}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \operatorname{rank}(\mathbf{A}) < \operatorname{rank}([\mathbf{A} \mid \mathbf{b}]) \text{ (parallel)} \tag{1.1.5.4}$$

where the ratios are read as cross-multiplied products if a denominator is zero. Solution: $\left|\mathbf{A}\right| = a_1b_2 - a_2b_1 \neq 0$ iff the rows (the normals) are linearly independent, which is $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. If the second row of $\mathbf{A}$ is λ times the first, then $\operatorname{rank}(\mathbf{A}) = 1$, and the last row of the augmented matrix after $R_2 \to R_2 - \lambda R_1$ is $\begin{pmatrix} 0 & 0 & b_2 - \lambda b_1 \end{pmatrix}$, which is zero iff the constants are in the same ratio.

1.1.6 (Uniform Sample Space) For Ω with equally likely outcomes and an event $E \subseteq \Omega$,

$$\Pr(E) = \frac{|E|}{|\Omega|}, \quad \Pr(E^c) = 1 - \Pr(E), \quad \Pr(A \mid B) = \frac{\Pr(AB)}{\Pr(B)} \tag{1.1.6.1}$$

The number of multiples of both a and b in $\{1, 2, \ldots, N\}$ is

$$\left\lfloor \frac{N}{\operatorname{lcm}(a, b)} \right\rfloor \tag{1.1.6.2}$$

and the number of outcomes when k dice are thrown is

$$\left[\left(x + x^2 + \cdots + x^6\right)^k\right]_{x=1} = 6^k \tag{1.1.6.3}$$

Solution: The coefficient of x^n in $\left(x+\cdots+x^6\right)^k$ is the number of outcomes of the k dice with sum n, since multiplying generating functions is the discrete convolution of the counts. Setting $x=1$ adds up the counts of all sums, giving (1.1.6.3). A number is a multiple of a and b iff it is a multiple of $\operatorname{lcm}(a,b)$, which gives (1.1.6.2).

1.1.7 (Grouped Data: Mode and Median) Let the classes be $[l_i, l_i+h]$ with frequencies f_i, $N=\sum_i f_i$ and cumulative frequency $F_i=\sum_{j\le i} f_j$. If $[l, l+h]$ is the modal class with frequency f_1, and f_0, f_2 are the frequencies of the preceding and succeeding classes,

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2}\, h \tag{1.1.7.1}$$

If $[l, l+h]$ is the class with $F_{k-1} < \frac{N}{2} \le F_k$ and frequency f_k,

$$\text{Median} = l + \frac{\frac{N}{2} - F_{k-1}}{f_k}\, h \tag{1.1.7.2}$$

Solution: On the histogram, join the top-right corner of the preceding bar to the top-right corner of the modal bar, and the top-left corner of the modal bar to the top-left corner of the succeeding bar. With $t=\frac{x-l}{h}$ the two lines are

$$y = f_0 + (f_1 - f_0)\, t, \qquad y = f_1 + (f_2 - f_1)\, t \tag{1.1.7.3}$$

Equating them, $t=\frac{f_1-f_0}{2f_1-f_0-f_2}$, and $x=l+th$ is (1.1.7.1). For the median, the ogive (cumulative frequency) is piecewise linear, and the median solves $F(x)=\frac{N}{2}$. On the class $[l, l+h]$, $F(x)=F_{k-1}+f_k\frac{x-l}{h}$, which gives (1.1.7.2).

1.1.8 (Incircle of a Triangle) Let a, b, c be the sides opposite $\mathbf{A}, \mathbf{B}, \mathbf{C}$, $s=\frac{a+b+c}{2}$ and Δ the area. The incentre $\mathbf{I}$ and inradius r are

$$\mathbf{I} = \frac{a\mathbf{A} + b\mathbf{B} + c\mathbf{C}}{a+b+c}, \qquad r = \frac{\Delta}{s} \tag{1.1.8.1}$$

For a right angle at $\mathbf{B}$ (legs a, c, hypotenuse b),

$$r = \frac{a + c - b}{2} \tag{1.1.8.2}$$

Equivalently, with $\mathbf{n}_i^{\top}\mathbf{x} = c_i$ the three sides in terms of inward unit normals, $(\mathbf{I}, r)$ solves the linear system

$$\mathbf{n}_i^{\top}\mathbf{I} - r = c_i, \quad i = 1,2,3 \tag{1.1.8.3}$$

Solution: Joining $\mathbf{I}$ to the vertices splits the triangle into three triangles of height r on the bases a, b, c, so $\Delta = \frac{1}{2}r\,(a+b+c) = rs$. For a right angle at $\mathbf{B}$ the two tangent segments from $\mathbf{B}$ equal r, and those from $\mathbf{A}$, $\mathbf{C}$ are $c-r$ and $a-r$, so $b=(c-r)+(a-r)$, giving (1.1.8.2). The signed distance of $\mathbf{I}$ from the line $\mathbf{n}_i^{\top}\mathbf{x}=c_i$ is $\mathbf{n}_i^{\top}\mathbf{I}-c_i=r$ for each side, giving (1.1.8.3).

1.1.9 (Polynomial Division and Remainder Theorem) For polynomials f and $g \neq 0$ there are unique q and r with

$$f(x) = q(x)\, g(x) + r(x), \quad \deg r < \deg g \tag{1.1.9.1}$$

If α is a (possibly complex) root of g, then

$$r(\alpha) = f(\alpha) \tag{1.1.9.2}$$

Solution: Writing the coefficients of q and r as unknowns, (1.1.9.1) is a linear system (a convolution matrix acting on the coefficients of q, plus the coefficients of r) with a unique solution. Substituting $x = \alpha$ with $g(\alpha) = 0$ gives (1.1.9.2). If $\deg g = 2$ with distinct roots α_1, α_2, then $r(x) = r_1 x + r_0$ is the line through $(\alpha_i, f(\alpha_i))$.

1.1.10 (Volume Flow Rate) If a fluid moves with uniform speed v normal to a cross-section S, the volume flow rate and the volume in time T are

$$Q = \iint_S v\, dA = v\pi R^2, \qquad V = \int_0^T Q\, dt = QT \tag{1.1.10.1}$$

Solution: For a circular cross-section of radius R, in polar coordinates

$$\iint_S v\, dA = \int_0^{2\pi} \int_0^R v\rho\, d\rho\, d\theta = v \cdot 2\pi \cdot \frac{R^2}{2} = v\pi R^2 \tag{1.1.10.2}$$

Also, 1 L = 1000 cm^3.

1.1.11 (Angle of Depression and Uniform Motion) An observer at height h above the ground sees an object at horizontal distance x at an angle of depression θ, where

$$\tan\theta = \frac{h}{x} \iff x = h\cot\theta \tag{1.1.11.1}$$

If the object moves towards the base with uniform speed v,

$$x(t) = x_0 - vt \tag{1.1.11.2}$$

Two observations $x(t_1) = x_1$, $x(t_2) = x_2$ give the linear system

$$\begin{pmatrix} 1 & -t_1 \\ 1 & -t_2 \end{pmatrix} \begin{pmatrix} x_0 \\ v \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \tag{1.1.11.3}$$

and the object reaches the base when $x(t) = 0$. Solution: The line of sight from the eye to the object makes a right triangle with the vertical of length h and the ground distance x, with the angle at the object equal to the angle of depression, giving (1.1.11.1).

B. Problems

1.2.1 PQ is a tangent to a circle with centre O at the point P on the circle. If $\triangle OPQ$ is an isosceles triangle, then find $\angle OQP$. (CBSE X 2021)

Solution: From (1.1.4.2), the direction $\mathbf{m} = \mathbf{Q} - \mathbf{P}$ of the tangent satisfies

$$(\mathbf{Q} - \mathbf{P})^{\top} (\mathbf{P} - \mathbf{O}) = 0 \implies \angle OPQ = 90^{\circ} \tag{1.2.1.1}$$

Hence, by Pythagoras,

$$\|\mathbf{O} - \mathbf{Q}\|^2 = \|\mathbf{O} - \mathbf{P}\|^2 + \|\mathbf{P} - \mathbf{Q}\|^2 \tag{1.2.1.2}$$

so OQ is strictly the longest side, and the equal sides of the isosceles triangle must be OP and PQ. Since the shape is fixed up to rotation and scaling, take

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix} r \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{Q} = \mathbf{P} + r\mathbf{m} = \begin{pmatrix} r \\ r \end{pmatrix} \tag{1.2.1.3}$$

Substituting in (1.1.4.1),

$$\cos \angle OQP = \frac{(\mathbf{O} - \mathbf{Q})^{\top} (\mathbf{P} - \mathbf{Q})}{\|\mathbf{O} - \mathbf{Q}\| \, \|\mathbf{P} - \mathbf{Q}\|} = \frac{\begin{pmatrix} -r & -r \end{pmatrix} \begin{pmatrix} 0 \\ -r \end{pmatrix}}{r\sqrt{2} \cdot r} = \frac{1}{\sqrt{2}} \tag{1.2.1.4}$$

$$\therefore \angle OQP = 45^{\circ} \tag{1.2.1.5}$$

The following code verifies this in Fig. 1.2.1.1.

codes/cbXq6.py

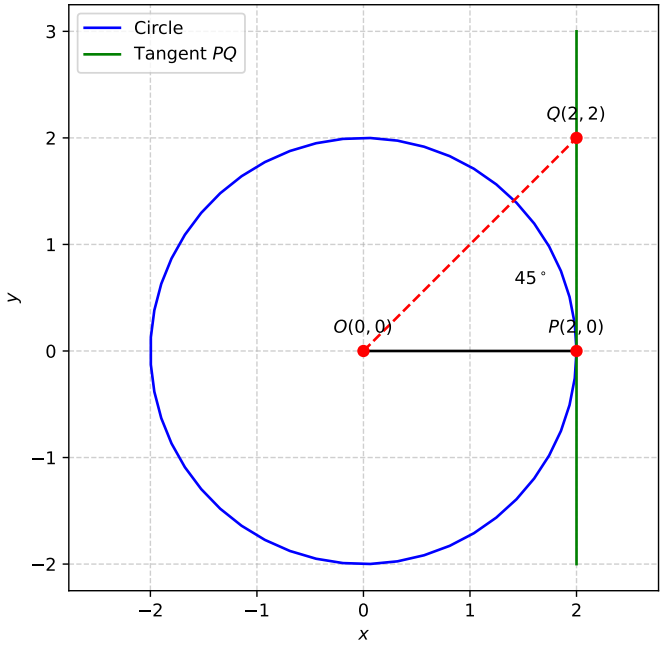


Fig. 1.2.1.1. Tangent PQ and the isosceles triangle OPQ with $\angle OQP = 45°$

1.2.2 Two concentric circles have radii 10 cm and 6 cm. Find the length of the chord of the larger circle which touches the smaller circle. (CBSE X 2021)
Solution: With the common centre at the origin, from (1.1.2.2) the circles have

$$\mathbf{V} = \mathbf{I}, \quad \mathbf{u} = \mathbf{0}, \quad f_1 = -100 \text{ (larger)}, \quad f_2 = -36 \text{ (smaller)} \tag{1.2.2.1}$$

Choose the axes so that the chord is the line $\mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$ with

$$\mathbf{h} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{m}^\top \mathbf{h} = 0 \tag{1.2.2.2}$$

For the smaller circle, $\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) = 0$ and $g(\mathbf{h}) = \|\mathbf{h}\|^2 - 36 = 0$, so (1.1.3.2) gives $\Delta = 0$ and the line is tangent. For the larger circle, $g(\mathbf{h}) = 36 - 100 = -64$, $\mathbf{m}^\top \mathbf{V}\mathbf{m} = 1$, so from (1.1.3.1)

$$\kappa = \frac{0 \pm \sqrt{0 + 64}}{1} = \pm 8 \implies \mathbf{x}_{1,2} = \begin{pmatrix} 6 \\ \pm 8 \end{pmatrix} \tag{1.2.2.3}$$

Hence the length of the chord, from (1.1.3.2), is

$$\|\mathbf{x}_1 - \mathbf{x}_2\| = |8 - (-8)| \, \|\mathbf{m}\| = 16 \text{ cm} \tag{1.2.2.4}$$

The following code verifies this in Fig. 1.2.2.1.

codes/cbXq7.py

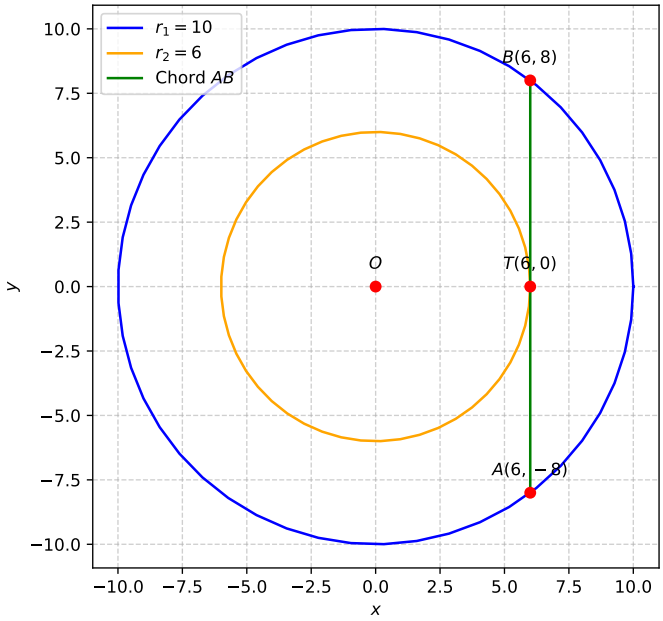


Fig. 1.2.2.1. Chord of the larger circle touching the smaller circle

1.2.3 How many outcomes are possible when three dice are thrown together? (CBSE X 2021)

Solution: For one die, the generating function of the outcomes is $x+x^2+\cdots+x^6$. For three dice, from (1.1.6.3), the number of outcomes with sum n is the coefficient of x^n in

$$\left(x + x^2 + \cdots + x^6\right)^3 = x^3 + 3x^4 + 6x^5 + 10x^6 + 15x^7 + 21x^8 \tag{1.2.3.1}$$

$$+ 25x^9 + 27x^{10} + \cdots + x^{18} \tag{1.2.3.2}$$

and setting $x = 1$ adds the counts of all sums,

$$(1 + 1 + \cdots + 1)^3 = 6^3 = 216 \tag{1.2.3.3}$$

The following code verifies this in Fig. 1.2.3.1.

codes/cbXq16.py

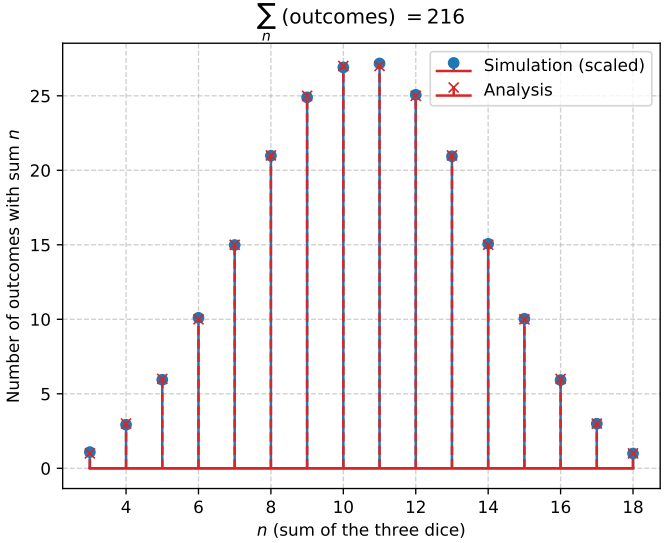


Fig. 1.2.3.1. Number of outcomes for each sum of three dice: Simulation vs Analysis

1.2.4 The residents of a housing society decided to build two straight paths in the central park. Taking one corner of the park as origin and two mutually perpendicular lines as the axes, the paths are represented by $2x - 3y = 5$ and $-6x + 9y = 7$. (CBSE X 2021)

(i) The two paths are

a) intersecting
b) overlapping
c) parallel
d) mutually perpendicular

(ii) Which of the following points lies on the line $2x - 3y = 5$?

a) $(-4, 1)$ b) $(4, -1)$ c) $(4, 1)$ d) $(-4, -1)$

(iii) If the line $-6x + 9y = 7$ intersects the y-axis at a point, its coordinates are

a) $\left(0, \frac{7}{9}\right)$ b) $\left(\frac{7}{9}, 0\right)$ c) $\left(-\frac{7}{6}, 0\right)$ d) $\left(0, -\frac{7}{6}\right)$

(iv) If a pair of equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ has a unique solution, then

a) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ b) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ c) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ d) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

(v) If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, the two lines are

a) parallel c) intersecting to each other
b) coincident d) perpendicular

Solution: The paths are the matrix equation (1.1.5.1) with

$$\mathbf{A} = \begin{pmatrix} 2 & -3 \\ -6 & 9 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 7 \end{pmatrix} \tag{1.2.4.1}$$

(i) The second row of $\mathbf{A}$ is -3 times the first, so $|\mathbf{A}| = 18 - 18 = 0$ and $\text{rank}\,(\mathbf{A}) = 1$. Applying $R_2 \to R_2 + 3R_1$ to the augmented matrix,

$$\left(\begin{array}{cc|c} 2 & -3 & 5 \\ -6 & 9 & 7 \end{array}\right) \xrightarrow{R_2 \to R_2 + 3R_1} \left(\begin{array}{cc|c} 2 & -3 & 5 \\ 0 & 0 & 22 \end{array}\right) \tag{1.2.4.2}$$

so $\text{rank}\,(\mathbf{A}) = 1 < 2 = \text{rank}\,([\mathbf{A} \mid \mathbf{b}])$. From (1.1.5.4) the system is inconsistent with parallel normals, so the paths are parallel: option (c). Their distance, writing the second line as $\mathbf{n}_1^\top \mathbf{x} = -\frac{7}{3}$ with $\mathbf{n}_1 = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$, is

$$\frac{\left|5 - \left(-\frac{7}{3}\right)\right|}{\|\mathbf{n}_1\|} = \frac{22}{3\sqrt{13}} \approx 2.03 \tag{1.2.4.3}$$

(ii) Evaluating $\mathbf{n}_1^\top \mathbf{x}_0$ for the four points gives $-11,\ 11,\ 5,\ -5$, and only $(4, 1)$ gives 5: option (c).
(iii) On the y-axis $\mathbf{e}_1^\top \mathbf{x} = 0$, so $9y = 7$ and the point is $\left(0, \frac{7}{9}\right)$: option (a).
(iv) A unique solution requires $\text{rank}\,(\mathbf{A}) = 2$, i.e. $|\mathbf{A}| = a_1 b_2 - a_2 b_1 \neq 0$. From (1.1.5.4) this is $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$: option (b).
(v) If the rows are proportional, $\text{rank}\,(\mathbf{A}) = \text{rank}\,([\mathbf{A} \mid \mathbf{b}]) = 1$, so the system has infinitely many solutions and the lines are coincident: option (b).
The following code verifies the results in Fig. 1.2.4.1.

codes/cbXq17.py

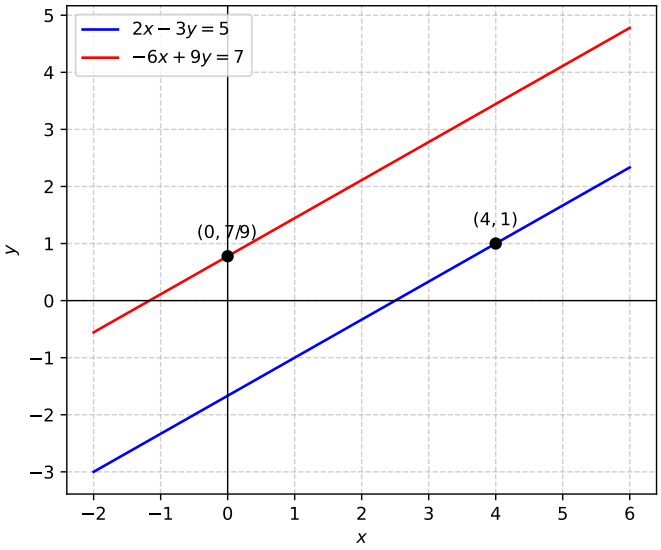


Fig. 1.2.4.1. The parallel paths $2x - 3y = 5$ and $-6x + 9y = 7$

1.2.5 The time taken (in seconds) by a group of students to complete a race is recorded as follows.

Time (s)	0–20	20–40	40–60	60–80	80–100
Students	1	4	3	7	5

(CBSE X 2021)

(i) The class mark of the modal class is

a) 60 b) 70 c) 80 d) 140

(ii) The mode of the given data is

a) 70.33 b) 71.33 c) 72.33 d) 73.33

(iii) The median class of the given data is

a) 20–40 b) 40–60 c) 80–100 d) 60–80

(iv) The sum of the lower limits of the median class and the modal class is

a) 80 b) 140 c) 120 d) 100

(v) The median time (in seconds) of the given data is

a) 65.7 b) 85.7 c) 45.7 d) 25.7

Solution: Here $h = 20$, $N = \sum_i f_i = 20$, and the cumulative frequencies are

$$F_i = 1,\ 5,\ 8,\ 15,\ 20 \tag{1.2.5.1}$$

(i) The largest frequency is 7, so the modal class is 60–80 with class mark $\frac{60+80}{2} = 70$: option (b).
(ii) With $l = 60$, $f_0 = 3$, $f_1 = 7$, $f_2 = 5$, the two lines of the histogram construction pass through

$$\mathbf{A}_1 = \begin{pmatrix} 60 \\ 3 \end{pmatrix},\ \mathbf{B}_1 = \begin{pmatrix} 80 \\ 7 \end{pmatrix},\qquad \mathbf{A}_2 = \begin{pmatrix} 60 \\ 7 \end{pmatrix},\ \mathbf{B}_2 = \begin{pmatrix} 80 \\ 5 \end{pmatrix} \tag{1.2.5.2}$$

The direction vectors are $\mathbf{m}_1 = \begin{pmatrix} 20 \\ 4 \end{pmatrix}$, $\mathbf{m}_2 = \begin{pmatrix} 20 \\ -2 \end{pmatrix}$, with normals obtained by a 90° rotation, $\mathbf{n}_i = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{m}_i$, so that

$$\begin{pmatrix} 4 & -20 \\ -2 & -20 \end{pmatrix} \mathbf{x} = \begin{pmatrix} \mathbf{n}_1^\top \mathbf{A}_1 \\ \mathbf{n}_2^\top \mathbf{A}_2 \end{pmatrix} = \begin{pmatrix} 180 \\ -260 \end{pmatrix} \implies \mathbf{x} = \begin{pmatrix} 73.33 \\ 5.67 \end{pmatrix} \tag{1.2.5.3}$$

Its abscissa is the mode, which agrees with (1.1.7.1)

$$60 + \frac{7-3}{14-3-5} \times 20 = 73.33 \tag{1.2.5.4}$$

option (d).
(iii) Since $\frac{N}{2} = 10$ and $F_3 = 8 < 10 \le 15 = F_4$, the median class is 60–80: option (d).
(iv) The lower limits of both classes are 60, so the sum is 120: option (c).
(v) From (1.1.7.2), with $l = 60$, $F_{k-1} = 8$, $f_k = 7$,

$$\text{Median} = 60 + \frac{10-8}{7} \times 20 = 60 + \frac{40}{7} \approx 65.7 \tag{1.2.5.5}$$

option (a). The following code verifies the mode as a line intersection and the median as the root of the ogive in Fig. 1.2.5.1.

codes/cbXq19.py

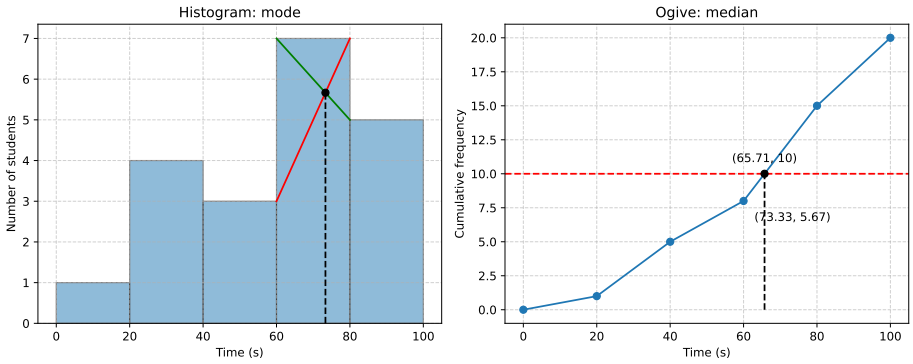


Fig. 1.2.5.1. Histogram construction for the mode and ogive for the median

1.2.6 Twenty identical cards are numbered 1 to 20 and placed in a box. One card is drawn at random. (CBSE X 2021)

(i) The probability that the number on the card is an odd prime number is

a) $\frac{3}{5}$ b) $\frac{2}{5}$ c) $\frac{9}{20}$ d) $\frac{7}{20}$

(ii) The probability that the number on the card is a composite number is

a) $\frac{11}{20}$ b) $\frac{3}{5}$ c) $\frac{4}{5}$ d) $\frac{1}{2}$

(iii) The probability that the number on the card is a multiple of 3, 6 and 9 is

a) $\frac{1}{20}$ b) $\frac{1}{10}$ c) $\frac{3}{20}$ d) 0

(iv) The probability that the number on the card is a multiple of 3 and 7 is

a) $\frac{3}{10}$ b) $\frac{1}{10}$ c) 0 d) $\frac{2}{5}$

(v) If all cards with odd numbers are removed and one card is drawn from the rest, the probability of a prime number is

a) $\frac{1}{20}$ b) $\frac{1}{10}$ c) 0 d) $\frac{1}{5}$

Solution: Let $\Omega = \{1, 2, \ldots, 20\}$ with $p_X(n) = \frac{1}{20}$ and use (1.1.6.1). The primes in Ω are

$$\{2, 3, 5, 7, 11, 13, 17, 19\}, \quad |\{\text{primes}\}| = 8 \tag{1.2.6.1}$$

(i) Removing 2 leaves 7 odd primes, so $\Pr(\text{odd prime}) = \frac{7}{20}$: option (d).
(ii) Neither 1 nor a prime is composite, so $|\{\text{composite}\}| = 20 - 1 - 8 = 11$ and $\Pr(\text{composite}) = \frac{11}{20}$: option (a).

(iii) A multiple of 3, 6 and 9 is a multiple of $\text{lcm}\,(3,6,9) = 18$. From (1.1.6.2), the count is $\left\lfloor \frac{20}{18} \right\rfloor = 1$, so the probability is $\frac{1}{20}$: option (a).
(iv) $\text{lcm}\,(3,7) = 21 > 20$, so the count is $\left\lfloor \frac{20}{21} \right\rfloor = 0$ and the probability is 0: option (c).
(v) Let E be the event "even number" and S the event "prime". Then $\Pr(E) = \frac{10}{20}$ and $SE = \{2\}$, so

$$\Pr(S \mid E) = \frac{\Pr(SE)}{\Pr(E)} = \frac{1/20}{1/2} = \frac{1}{10} \tag{1.2.6.2}$$

option (b). The following code verifies these in Fig. 1.2.6.1.

codes/cbXq20.py

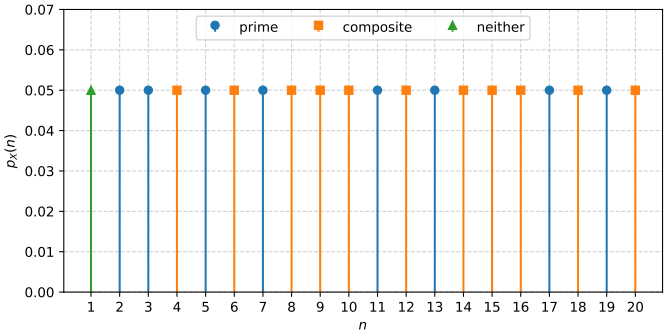


Fig. 1.2.6.1. PMF of the card drawn, grouped as prime, composite and neither

1.2.7 Find the value of k for which the system of equations $x + 2y = 5$ and $3x + ky + 15 = 0$ has no solution. (CBSE X 2021)
Solution: From (1.1.5.1),

$$\begin{pmatrix} 1 & 2 \\ 3 & k \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ -15 \end{pmatrix} \tag{1.2.7.1}$$

Applying Gaussian elimination to the augmented matrix,

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & k & -15 \end{array}\right) \xrightarrow{R_2 \to R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & k-6 & -30 \end{array}\right) \tag{1.2.7.2}$$

If $k \neq 6$, then $\text{rank}\,(\mathbf{A}) = 2$ and the solution is unique. If $k = 6$, then $\text{rank}\,(\mathbf{A}) = 1$ while the last row of the augmented matrix is $\begin{pmatrix} 0 & 0 & -30 \end{pmatrix}$, so $\text{rank}\,([\mathbf{A} \mid \mathbf{b}]) = 2$. From (1.1.5.4),

$$\text{rank}\,(\mathbf{A}) < \text{rank}\,([\mathbf{A} \mid \mathbf{b}]) \implies k = 6 \tag{1.2.7.3}$$

This agrees with $\frac{1}{3} = \frac{2}{k} \neq \frac{-5}{15}$ at $k = 6$. The following code verifies this in Fig. 1.2.7.1.

codes/cbXq22.py

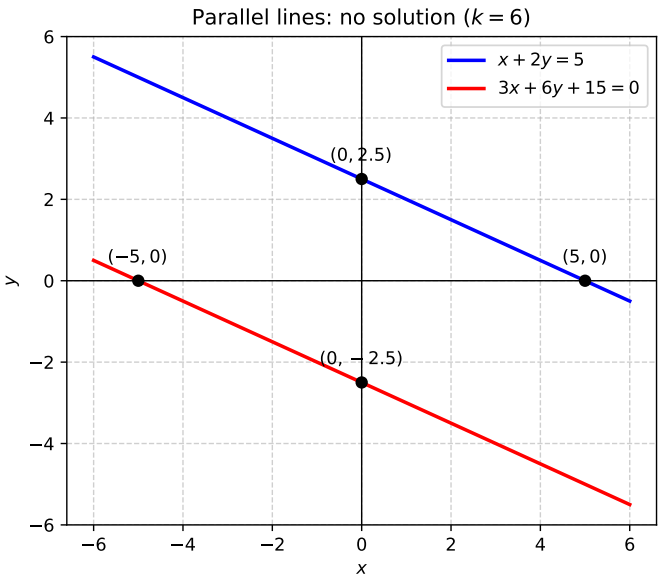


Fig. 1.2.7.1. Parallel lines $x + 2y = 5$ and $3x + 6y + 15 = 0$ ($k = 6$): no solution

1.2.8 ABC is a right triangle, right-angled at B, with $BC = 6$ cm and $AB = 8$ cm. A circle with centre O and radius r cm is inscribed in $\triangle ABC$. Find the value of r. (CBSE X 2021)

Solution: Taking B as the origin with BC and BA along the axes,

$$\mathbf{A} = \begin{pmatrix} 0 \\ 8 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \tag{1.2.8.1}$$

The sides are $a = BC = 6$, $c = AB = 8$ and $b = CA = \|\mathbf{C} - \mathbf{A}\| = 10$. From (1.1.8.1),

$$\mathbf{I} = \frac{6\begin{pmatrix} 0 \\ 8 \end{pmatrix} + 10\begin{pmatrix} 0 \\ 0 \end{pmatrix} + 8\begin{pmatrix} 6 \\ 0 \end{pmatrix}}{24} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \qquad r = \frac{\Delta}{s} = \frac{\frac{1}{2} \cdot 6 \cdot 8}{12} = 2 \tag{1.2.8.2}$$

Using the linear system (1.1.8.3) with the inward unit normals of AB, BC and CA ($4x + 3y = 24$),

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ -0.8 & -0.6 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ r \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -4.8 \end{pmatrix} \implies \begin{pmatrix} x \\ y \\ r \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \tag{1.2.8.3}$$

Also, from (1.1.8.2), $r = \frac{6+8-10}{2} = 2$.

$$\therefore r = 2 \text{ cm} \tag{1.2.8.4}$$

The points of contact are $\mathbf{P} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$, $\mathbf{Q} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ and $\mathbf{R} = \begin{pmatrix} 3.6 \\ 3.2 \end{pmatrix}$. The following code verifies this in Fig. 1.2.8.1.

codes/cbXq25.py

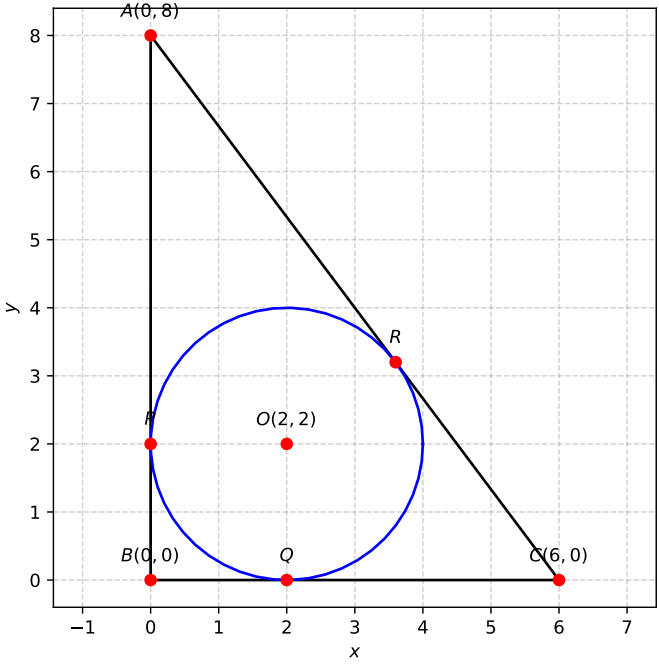


Fig. 1.2.8.1. Incircle of the right triangle ABC with $r = 2$

1.2.9 Divide the polynomial $f(x) = 5x^3 + 10x^2 - 30x - 15$ by the polynomial $g(x) = x^2 + 1 + x$ and hence find the quotient and the remainder. (CBSE X 2021)
Solution: From (1.1.9.1), let $q(x) = q_1 x + q_0$ and $r(x) = r_1 x + r_0$. Then

$$q(x)\, g(x) + r(x) = q_1 x^3 + (q_1 + q_0)\, x^2 + (q_1 + q_0 + r_1)\, x + (q_0 + r_0) \tag{1.2.9.1}$$

Comparing coefficients with $f(x)$ gives the matrix equation

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} q_1 \\ q_0 \\ r_1 \\ r_0 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \\ -30 \\ -15 \end{pmatrix} \implies \begin{pmatrix} q_1 \\ q_0 \\ r_1 \\ r_0 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \\ -40 \\ -20 \end{pmatrix} \tag{1.2.9.2}$$

Verifying the remainder with (1.1.9.2): the roots of g are ω, ω^2 with $\omega^3 = 1$ and $\omega^2 = -1 - \omega$, so

$$f(\omega) = 5 + 10\omega^2 - 30\omega - 15 = -20 - 40\omega = r(\omega) \tag{1.2.9.3}$$

and similarly for ω^2, which gives $r(x) = -40x - 20$.

$$\therefore q(x) = 5x + 5, \quad r(x) = -40x - 20 \tag{1.2.9.4}$$

The following code verifies this in Fig. 1.2.9.1.

codes/cbXq27.py

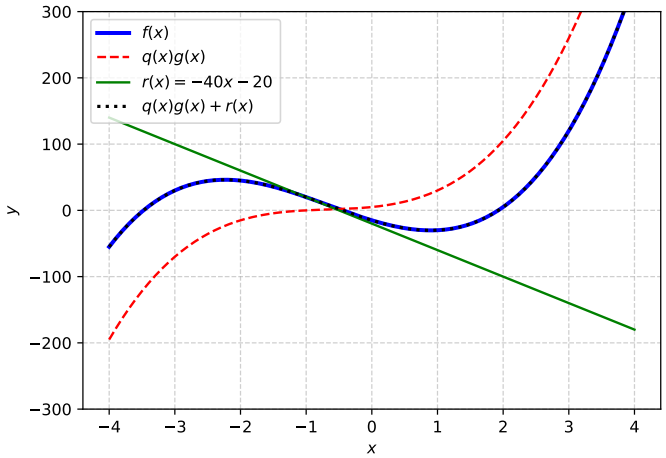


Fig. 1.2.9.1. $f(x)$ and $q(x)\,g(x) + r(x)$ coincide

1.2.10 Water is being pumped out through a circular pipe whose internal diameter is 8 cm. If the rate of flow of water is 80 cm/s, how many litres of water are pumped out through this pipe in one hour? (CBSE X 2021)

Solution: Here $R = 4$ cm, $v = 80$ cm/s and $T = 3600$ s. From (1.1.10.1),

$$Q = \int_0^{2\pi} \int_0^4 80\rho \, d\rho \, d\theta = 80 \cdot 2\pi \cdot 8 = 1280\pi \text{ cm}^3/\text{s} \tag{1.2.10.1}$$

$$V = \int_0^{3600} Q \, dt = 1280\pi \times 3600 = 4608000\pi \text{ cm}^3 \tag{1.2.10.2}$$

Since 1 L $= 1000$ cm^3,

$$V = 4608\pi \text{ L} \approx 14476.46 \text{ L} \tag{1.2.10.3}$$

(with $\pi = \frac{22}{7}$, $V \approx 14482.29$ L). The following code verifies this in Fig. 1.2.10.1.

codes/cbXq34.py

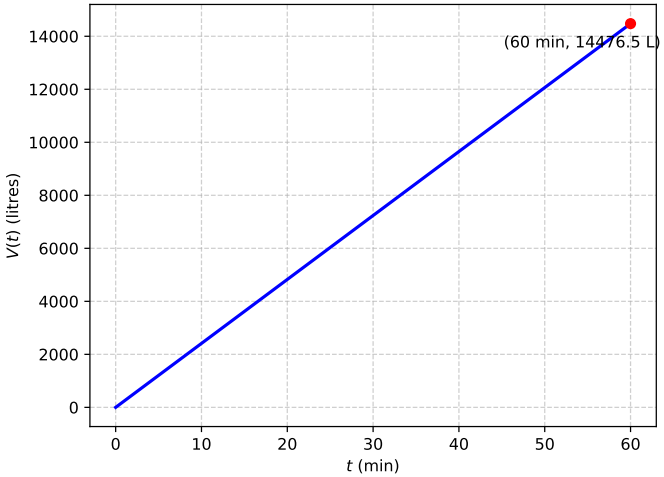


Fig. 1.2.10.1. Volume pumped out versus time

1.2.11 A man on the top of a vertical tower observes a car moving at a uniform speed coming directly towards it. If it takes 18 minutes for the angle of depression to change from 30° to 60°, how soon after this will the car reach the tower? (CBSE X 2021)

Solution: Let the tower have height h and let $x(t)$ be the horizontal distance of the car from its foot, with $t = 0$ at the first observation. From (1.1.11.1),

$$x(0) = h\cot 30^\circ = \sqrt{3}h, \qquad x(18) = h\cot 60^\circ = \frac{h}{\sqrt{3}} \tag{1.2.11.1}$$

Using (1.1.11.2) and (1.1.11.3) with $t_1 = 0$, $t_2 = 18$,

$$\begin{pmatrix} 1 & 0 \\ 1 & -18 \end{pmatrix} \begin{pmatrix} x_0 \\ v \end{pmatrix} = \begin{pmatrix} \sqrt{3}h \\ \frac{h}{\sqrt{3}} \end{pmatrix} \implies x_0 = \sqrt{3}h, \quad v = \frac{1}{18}\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)h = \frac{h}{9\sqrt{3}} \tag{1.2.11.2}$$

The car reaches the tower when $x(t) = 0$, that is after a further time

$$t^* = \frac{x(18)}{v} = \frac{h/\sqrt{3}}{h/\left(9\sqrt{3}\right)} = 9 \text{ minutes} \tag{1.2.11.3}$$

which is independent of h. The following code verifies this in Fig. 1.2.11.1.

codes/cbXq35.py

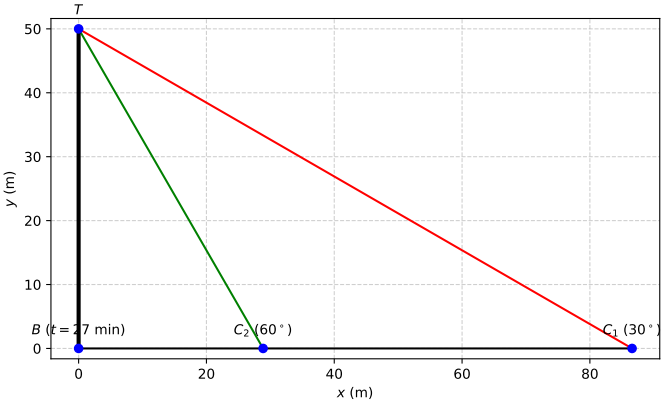


Fig. 1.2.11.1. Car approaching the tower: angles of depression 30° and 60°

2. CBSE XII 2021

A. Formulae

2.1.1 (Independent Events and Sum of Random Variables) If X and Y are independent, that is, the events $\{X = x\}$ and $\{Y = y\}$ are independent for all x, y, with PMFs p_X and p_Y,

$$p_{X,Y}(x,y) = p_X(x)\,p_Y(y), \qquad p_{X+Y}(n) = \sum_k p_X(k)\,p_Y(n-k) \tag{2.1.1.1}$$

and for independent events E_i with $\Pr(E_i) = p_i$,

$$\Pr(\text{at least one } E_i \text{ occurs}) = 1 - \prod_i (1 - p_i) \tag{2.1.1.2}$$

Solution: Independence of the events $\{X = x\}$ and $\{Y = y\}$ factorises the joint PMF. The event $\{X + Y = n\}$ is the disjoint union of $\{X = k,\, Y = n - k\}$ over k, which gives the convolution. At least one event occurs iff the sum of the indicators is nonzero, and $\Pr(\text{none}) = \prod_i (1 - p_i)$.

2.1.2 (Binomial Distribution) The number X of successes in n independent trials with success probability p has

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$
$$E[X] = np, \quad \operatorname{var}(X) = np(1-p) \tag{2.1.2.1}$$

Solution: One trial has the PMF $(1 - p,\ p)$, whose generating function is $1 - p + px$. By (2.1.1.1) the PMF of the sum of n trials is the n-fold convolution, that is, the coefficients of $(1 - p + px)^n$, which are $\binom{n}{k} p^k (1-p)^{n-k}$ by the binomial theorem. The mean and variance add over the independent trials.

2.1.3 (Hypergeometric Distribution) Draw n items without replacement from M items, of which K are of one kind. The number X of items of that kind has

$$p_X(k) = \frac{\binom{K}{k}\binom{M-K}{n-k}}{\binom{M}{n}}, \qquad E[X] = \frac{nK}{M} \tag{2.1.3.1}$$

Solution: All $\binom{M}{n}$ subsets of size n are equally likely, and those with exactly k items of the first kind can be chosen in $\binom{K}{k}\binom{M-K}{n-k}$ ways. Writing $X = \sum_{i=1}^{n} X_i$, where X_i indicates that the i-th draw is of the first kind, symmetry gives $\Pr(X_i = 1) = \frac{K}{M}$ and hence the mean.

2.1.4 (Total Probability and Bayes' Theorem) If $B_1, \ldots, B_m$ partition the sample space and E is an event,

$$\Pr(B_j \mid E) = \frac{\Pr(E \mid B_j)\Pr(B_j)}{\sum_i \Pr(E \mid B_i)\Pr(B_i)} \tag{2.1.4.1}$$

Solution: $\Pr(B_j E) = \Pr(E \mid B_j)\Pr(B_j) = \Pr(B_j \mid E)\Pr(E)$, and $\Pr(E) = \sum_i \Pr(EB_i) = \sum_i \Pr(E \mid B_i)\Pr(B_i)$.

2.1.5 (Lines in Space) A line through $\mathbf{a}$ with direction $\mathbf{m}$ is $\mathbf{x} = \mathbf{a} + \kappa\mathbf{m}$. The symmetric form

$$\frac{x - x_0}{d_1} = \frac{y - y_0}{d_2} = \frac{z - z_0}{d_3} \implies \mathbf{m} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix} \tag{2.1.5.1}$$

A condition such as $x = 5$ means that coordinate is constant, so that component of $\mathbf{m}$ is 0. The angle θ between two lines is the angle between their directions, and they are perpendicular iff

$$\cos\theta = \frac{\mathbf{m}_1^\top \mathbf{m}_2}{\|\mathbf{m}_1\|\,\|\mathbf{m}_2\|} = 0 \iff \mathbf{m}_1^\top \mathbf{m}_2 = 0 \tag{2.1.5.2}$$

Solution: In the symmetric form, the point $(x_0 + \kappa d_1,\ y_0 + \kappa d_2,\ z_0 + \kappa d_3)$ moves along $\mathbf{m}$ as κ varies, so $\mathbf{m}$ is the direction. A constant coordinate does not change with κ, so the corresponding component is 0.

2.1.6 (Cross Product, Area and Collinearity) $\|\mathbf{u} \times \mathbf{v}\|$ is the area of the parallelogram on $\mathbf{u}, \mathbf{v}$, and $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ iff $\mathbf{u} \parallel \mathbf{v}$. For three points,

$$(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) = \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} + \mathbf{a} \times \mathbf{b} \tag{2.1.6.1}$$

The area of the triangle is half the norm of this vector, and the points are collinear iff it is $\mathbf{0}$. Solution: By bilinearity,

$$(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) = \mathbf{b} \times \mathbf{c} - \mathbf{b} \times \mathbf{a} - \mathbf{a} \times \mathbf{c} + \mathbf{a} \times \mathbf{a} \tag{2.1.6.2}$$

and $\mathbf{a} \times \mathbf{a} = \mathbf{0}$, $-\mathbf{b} \times \mathbf{a} = \mathbf{a} \times \mathbf{b}$, $-\mathbf{a} \times \mathbf{c} = \mathbf{c} \times \mathbf{a}$.

2.1.7 (Determinants: Line and Area) The line through (x_1, y_1) and (x_2, y_2), and the area of a triangle with vertices (x_i, y_i), are, respectively,

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0, \qquad \Delta = \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right| \tag{2.1.7.1}$$

Solution: The first determinant is affine in (x, y) and vanishes at both given points (two equal rows), so it is the equation of the line through them. For the second, subtracting row 3 from rows 1 and 2 and expanding along the last column,

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \begin{vmatrix} x_1 - x_3 & y_1 - y_3 \\ x_2 - x_3 & y_2 - y_3 \end{vmatrix} \tag{2.1.7.2}$$

which is the signed area of the parallelogram on $\mathbf{x}_1 - \mathbf{x}_3$ and $\mathbf{x}_2 - \mathbf{x}_3$, twice

the area of the triangle.

2.1.8 (Shortest Distance between Two Lines) For $\mathbf{x} = \mathbf{a}_1 + s\mathbf{m}_1$ and $\mathbf{x} = \mathbf{a}_2 + t\mathbf{m}_2$, let $\mathbf{M} = \begin{pmatrix} \mathbf{m}_1 & \mathbf{m}_2 \end{pmatrix}$ and $\mathbf{b} = \mathbf{a}_2 - \mathbf{a}_1$. Then

$$d = \min_{\mathbf{z}} \|\mathbf{M}\mathbf{z} - \mathbf{b}\|, \quad \mathbf{M}^\top \mathbf{M}\mathbf{z} = \mathbf{M}^\top \mathbf{b}, \quad \mathbf{z} = \begin{pmatrix} s \\ -t \end{pmatrix} \tag{2.1.8.1}$$

which is equivalent to

$$d = \frac{\left|\mathbf{b}^\top (\mathbf{m}_1 \times \mathbf{m}_2)\right|}{\|\mathbf{m}_1 \times \mathbf{m}_2\|} \tag{2.1.8.2}$$

The lines intersect iff $d = 0$. Solution: The distance between $\mathbf{a}_1 + s\mathbf{m}_1$ and $\mathbf{a}_2 + t\mathbf{m}_2$ is $\|s\mathbf{m}_1 - t\mathbf{m}_2 - \mathbf{b}\| = \|\mathbf{M}\mathbf{z} - \mathbf{b}\|$. It is minimum when the residual is orthogonal to both columns of $\mathbf{M}$ (the normal equations). The residual is then perpendicular to $\mathbf{m}_1$ and $\mathbf{m}_2$, so it is along $\mathbf{m}_1 \times \mathbf{m}_2$ and equals the projection of $\mathbf{b}$ on that direction, which is (2.1.8.2).

2.1.9 (Planes through a Line of Intersection) The planes $\mathbf{n}_1^\top \mathbf{x} + e_1 = 0$ and $\mathbf{n}_2^\top \mathbf{x} + e_2 = 0$ meet in a line, and every plane through that line is

$$(\mathbf{n}_1 + \lambda \mathbf{n}_2)^\top \mathbf{x} + (e_1 + \lambda e_2) = 0 \tag{2.1.9.1}$$

The distance of $\mathbf{n}^\top \mathbf{x} + e = 0$ from the origin is

$$\frac{|e|}{\|\mathbf{n}\|} \tag{2.1.9.2}$$

Solution: A point on both planes satisfies (2.1.9.1) for every λ. The point of the plane nearest to the origin lies along the normal, $\mathbf{x}^* = -\frac{e}{\|\mathbf{n}\|^2}\mathbf{n}$ (it satisfies $\mathbf{n}^\top \mathbf{x}^* + e = 0$), and $\|\mathbf{x}^*\| = \frac{|e|}{\|\mathbf{n}\|}$.

2.1.10 (Symmetric and Skew-Symmetric Parts) Every square matrix is $\mathbf{A} = \mathbf{S} + \mathbf{K}$ with

$$\mathbf{S} = \frac{\mathbf{A} + \mathbf{A}^\top}{2}, \qquad \mathbf{K} = \frac{\mathbf{A} - \mathbf{A}^\top}{2} \tag{2.1.10.1}$$

and $\mathbf{A}$ is symmetric iff $\mathbf{K} = \mathbf{0}$. Solution: $\mathbf{S}^\top = \mathbf{S}$, $\mathbf{K}^\top = -\mathbf{K}$ and $\mathbf{S} + \mathbf{K} = \mathbf{A}$. If $\mathbf{A} = \mathbf{A}^\top$, then $\mathbf{K} = \mathbf{0}$, and conversely.

2.1.11 (Idempotent Matrices and Matrix Polynomials) Let $\mathbf{A}^2 = \mathbf{A}$ and let p be a polynomial. Dividing $p(\lambda) = q(\lambda)\left(\lambda^2 - \lambda\right) + r_1\lambda + r_0$,

$$p(\mathbf{A}) = r_1\mathbf{A} + r_0\mathbf{I}, \qquad r_0 = p(0), \quad r_1 = p(1) - p(0) \tag{2.1.11.1}$$

Solution: Since $\mathbf{A}^2 - \mathbf{A} = \mathbf{0}$, the term $q(\mathbf{A})\left(\mathbf{A}^2 - \mathbf{A}\right)$ vanishes. Putting $\lambda = 0$ and $\lambda = 1$ (the roots of $\lambda^2 - \lambda$) gives $p(0) = r_0$ and $p(1) = r_1 + r_0$. Also, if $\mathbf{A}\mathbf{v} = \mu\mathbf{v}$ then $\mu^2 = \mu$, so the eigenvalues of $\mathbf{A}$ are 0 or 1.

2.1.12 (Relations as Boolean Matrices) On a finite set $\{x_1, \ldots, x_n\}$ let $M_{ij} = 1$ iff $x_i R x_j$. Then

$$\begin{aligned} \text{reflexive} &\iff M_{ii} = 1 \ \forall i \\ \text{symmetric} &\iff \mathbf{M} = \mathbf{M}^\top \\ \text{transitive} &\iff \left(\mathbf{M}^2 > 0\right) \subseteq (\mathbf{M} > 0) \end{aligned} \tag{2.1.12.1}$$

If $L_{ik} = 1$ when word i contains letter k, then "have a letter in common" is $\mathbf{M} = \left(\mathbf{L}\mathbf{L}^\top > 0\right)$. Solution: $\left(\mathbf{M}^2\right)_{ik} = \sum_j M_{ij} M_{jk}$ counts the j with $x_i R x_j R x_k$, so it is positive iff a two-step chain exists, and transitivity requires $x_i R x_k$ for every such chain. Also, $\left(\mathbf{L}\mathbf{L}^\top\right)_{ij} = \sum_k L_{ik} L_{jk}$ is the number of common letters, and it is symmetric by construction.

2.1.13 (Adjugate and Cayley-Hamilton) For a 2×2 matrix $\mathbf{A}$, the Cayley-Hamilton theorem states

$$\mathbf{A}^2 - \operatorname{tr}(\mathbf{A})\,\mathbf{A} + \left|\mathbf{A}\right|\mathbf{I} = \mathbf{0} \tag{2.1.13.1}$$

and the adjugate satisfies

$$\operatorname{adj}(\mathbf{A}) = \operatorname{tr}(\mathbf{A})\,\mathbf{I} - \mathbf{A}, \qquad \mathbf{A}\operatorname{adj}(\mathbf{A}) = \operatorname{adj}(\mathbf{A})\ \mathbf{A} = \left|\mathbf{A}\right|\mathbf{I} \tag{2.1.13.2}$$

Solution: For $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the transposed cofactor matrix is $\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (a+d)\,\mathbf{I} - \mathbf{A}$. By (2.1.13.1), $\mathbf{A}\left(\operatorname{tr}(\mathbf{A})\,\mathbf{I} - \mathbf{A}\right) = \operatorname{tr}(\mathbf{A})\,\mathbf{A} - \mathbf{A}^2 = \left|\mathbf{A}\right|\mathbf{I}$, and the other product is the same because $\mathbf{A}$ commutes with $\mathbf{I}$ and with itself. No inverse is used, so this holds even when $\left|\mathbf{A}\right| = 0$.

2.1.14 (Improper Integral and Substitution) If f is continuous on $[a, b]$ except for an unbounded point at b,

$$\int_a^b f(x)\,dx = \lim_{c \to b^-} \int_a^c f(x)\,dx \tag{2.1.14.1}$$

whenever the limit exists. With $x = \sin\theta$, a right triangle with hypotenuse 1 gives $\tan\left(\sin^{-1} x\right) = \frac{x}{\sqrt{1-x^2}}$ and $dx = \cos\theta\,d\theta$.

2.1.15 (Integration by Parts) $\int u\,dv = uv - \int v\,du$, and

$$\int e^x \left(f(x) + f'(x)\right) dx = e^x f(x) + c \tag{2.1.15.1}$$

Solution: With $u = f$ and $dv = e^x dx$, $\int e^x f\,dx = e^x f - \int e^x f'\,dx$. Adding $\int e^x f'\,dx$ to both sides gives (2.1.15.1).

2.1.16 (First-Order Linear ODE) The equation $\frac{dy}{dx} + P(x)\,y = Q(x)$ has the integrating

factor

$$\mu(x) = e^{\int P\,dx}, \qquad \frac{d}{dx}(\mu y) = \mu Q \tag{2.1.16.1}$$

Solution: By the product rule $\frac{d}{dx}(\mu y) = \mu\frac{dy}{dx} + \mu' y$, which equals $\mu\left(\frac{dy}{dx} + Py\right)$ iff $\mu' = P\mu$, that is $\mu = e^{\int P\,dx}$.

2.1.17 (Area between Curves) The area between $y = f_1(x)$ (above) and $y = f_2(x)$ on $[a, b]$ is $\int_a^b (f_1 - f_2)\,dx$. For the region between $x = y^2$ and $x = 4$, which is symmetric about the x-axis ($y = \pm\sqrt{x}$), the part with $x \le a$ has area

$$A(a) = \int_0^a 2\sqrt{x}\,dx = \frac{4}{3}a^{3/2} \tag{2.1.17.1}$$

and the line $x = a$ divides the region into two equal parts iff $A(a) = \frac{1}{2}A(4)$.

2.1.18 (Partial Fractions) A proper rational function with a repeated factor is

$$\frac{p(x)}{(x-r)^2(x-s)} = \frac{A}{x-r} + \frac{B}{(x-r)^2} + \frac{C}{x-s} \tag{2.1.18.1}$$

where A, B, C solve the linear system obtained by matching the coefficients of $p(x) = A(x-r)(x-s) + B(x-s) + C(x-r)^2$. Then $\int \frac{dx}{x-r} = \ln|x-r|$ and $\int \frac{dx}{(x-r)^2} = -\frac{1}{x-r}$.

B. Problems

2.2.1 The probabilities of solving a specific question independently by A and B are $\frac{1}{3}$ and $\frac{1}{5}$ respectively. If both try to solve the question independently, the probability that the question is solved is

a) $\frac{7}{15}$ b) $\frac{8}{15}$ c) $\frac{2}{15}$ d) $\frac{14}{15}$

(CBSE XII 2021)

Solution: Let $X_A, X_B \in \{0,1\}$ indicate that A and B solve the question, with PMFs

$$\left(p_A(0),\ p_A(1)\right) = \left(\frac{2}{3},\ \frac{1}{3}\right), \qquad \left(p_B(0),\ p_B(1)\right) = \left(\frac{4}{5},\ \frac{1}{5}\right) \tag{2.2.1.1}$$

From (2.1.1.1), the PMF of $S = X_A + X_B$ is the convolution

$$\left(p_S(0),\ p_S(1),\ p_S(2)\right) = \left(\frac{8}{15},\ \frac{6}{15},\ \frac{1}{15}\right) \tag{2.2.1.2}$$

where $p_S(1) = \frac{2}{3}\cdot\frac{1}{5} + \frac{1}{3}\cdot\frac{4}{5} = \frac{6}{15}$. The question is solved iff $S \geq 1$, so

$$\Pr(S \geq 1) = 1 - p_S(0) = 1 - \frac{8}{15} = \frac{7}{15} \tag{2.2.1.3}$$

which agrees with (2.1.1.2), $1 - \left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right)$. Hence the answer is option (a). The following code verifies this in Fig. 2.2.1.1.

codes/cbXIIq1.py

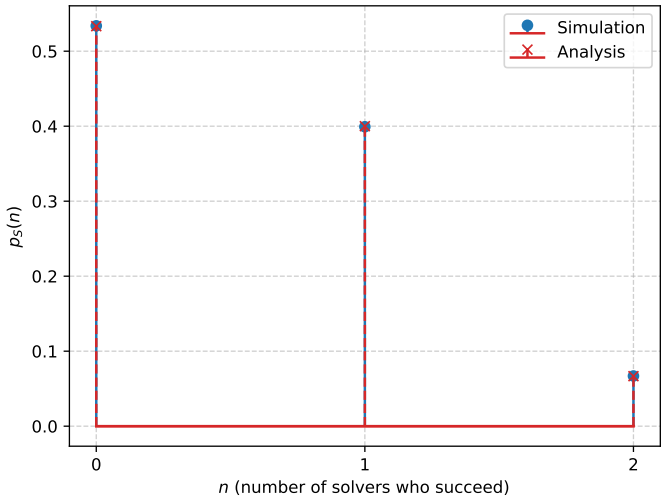


Fig. 2.2.1.1. PMF of the number of solvers who succeed: Simulation vs Analysis

2.2.2 If the two lines $L_1 : x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$ and $L_2 : x = 2, \frac{y}{-1} = \frac{z}{2-\alpha}$ are perpendicular, then the value of α is

a) $\frac{2}{3}$ b) 3 c) 4 d) $\frac{7}{3}$

(CBSE XII 2021)

Solution: From (2.1.5.1), the directions of the lines are (the condition $x = 5$ or $x = 2$ means the x component is 0)

$$\mathbf{m}_1 = \begin{pmatrix} 0 \\ 3-\alpha \\ -2 \end{pmatrix}, \qquad \mathbf{m}_2 = \begin{pmatrix} 0 \\ -1 \\ 2-\alpha \end{pmatrix} \tag{2.2.2.1}$$

From (2.1.5.2), the lines are perpendicular iff

$$\mathbf{m}_1^\top \mathbf{m}_2 = -(3-\alpha) - 2(2-\alpha) = 3\alpha - 7 = 0 \implies \alpha = \frac{7}{3} \tag{2.2.2.2}$$

For $\alpha = \frac{2}{3}, 3, 4$ the inner product is $-5, 2, 5$, all nonzero. Hence the answer is option (d). The following code verifies this in Fig. 2.2.2.1.

codes/cbXIIq3.py

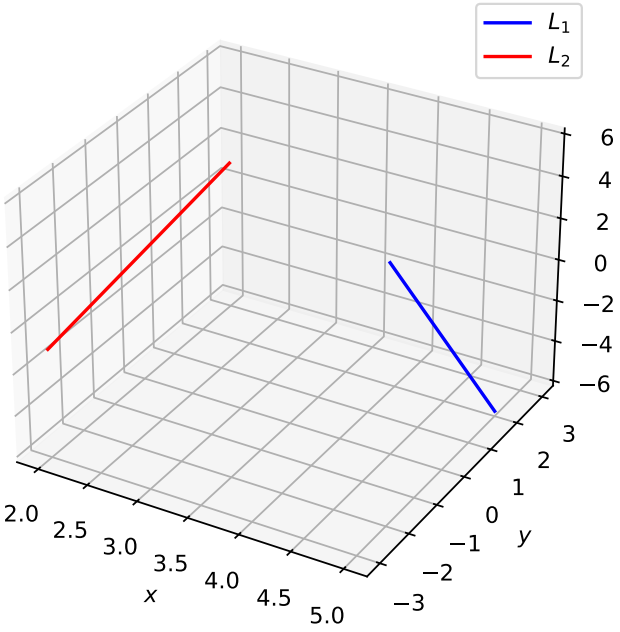


Fig. 2.2.2.1. The perpendicular lines L_1 and L_2 for $\alpha = \frac{7}{3}$

2.2.3 $\int_0^1 \tan\left(\sin^{-1} x\right) dx$ equals

a) 2 b) 0 c) -1 d) 1

(CBSE XII 2021)

Solution: From (2.1.14.1), $\tan\left(\sin^{-1} x\right) = \frac{x}{\sqrt{1-x^2}}$, which is unbounded as $x \to 1^-$, so the integral is improper. For $b < 1$,

$$\int_0^b \frac{x}{\sqrt{1-x^2}} dx = \left[-\sqrt{1-x^2}\right]_0^b = 1 - \sqrt{1-b^2} \tag{2.2.3.1}$$

and the limit is

$$\lim_{b\to 1^-} \left(1 - \sqrt{1-b^2}\right) = 1 \tag{2.2.3.2}$$

The substitution $x = \sin\theta$ gives the same value, $\int_0^{\pi/2} \tan\theta \cos\theta \, d\theta = \int_0^{\pi/2} \sin\theta \, d\theta = 1$. Hence the answer is option (d). The following code verifies this in Fig. 2.2.3.1.

codes/cbXIIq6.py

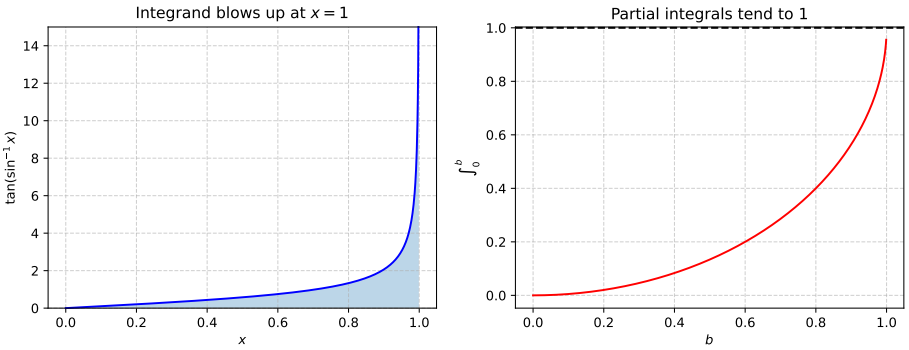


Fig. 2.2.3.1. The integrand and the partial integrals $\int_0^b$ tending to 1

2.2.4 $\int \frac{e^x}{x+1} \left[1 + (x+1)\log(x+1)\right] dx$ equals

a) $\frac{e^x}{x+1} + c$ b) $e^x \frac{x}{x+1} + c$ c) $e^x \log(x+1)$ d) $e^x \log(x+1) + c$ $e^x + c$

(CBSE XII 2021)

Solution: The integrand is

$$\frac{e^x}{x+1}\left[1 + (x+1)\log(x+1)\right] = e^x \left[\log(x+1) + \frac{1}{x+1}\right] \tag{2.2.4.1}$$

$$= e^x \left(f + f'\right), \quad f(x) = \log(x+1) \tag{2.2.4.2}$$

From (2.1.15.1),

$$\int e^x \left(f + f'\right) dx = e^x f + c = e^x \log\left(x + 1\right) + c \tag{2.2.4.3}$$

Hence the answer is option (d). Numerically, only option (d) has $F\left(b\right) - F\left(0\right)$ equal to $\int_0^b$ of the integrand (for $b = 1$ both are 1.8842). The following code verifies this in Fig. 2.2.4.1.

codes/cbXIIq7.py

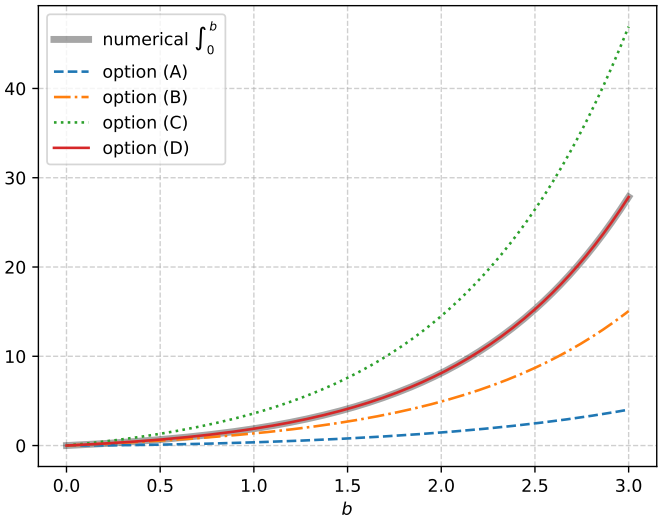


Fig. 2.2.4.1. Numerical $\int_0^b$ compared with the increments of the four options

2.2.5 The integrating factor of the differential equation $x\frac{dy}{dx} - y = \log x$ is _____. (CBSE XII 2021)

Solution: In the form $\frac{dy}{dx} + Py = Q$ we have $P = -\frac{1}{x}$ and $Q = \frac{\log x}{x}$. From (2.1.16.1),

$$\mu\left(x\right) = e^{\int P\,dx} = e^{-\log x} = \frac{1}{x} \tag{2.2.5.1}$$

Then $\frac{d}{dx}\left(\frac{y}{x}\right) = \frac{\log x}{x^2}$, and integrating by parts,

$$\frac{y}{x} = -\frac{\log x + 1}{x} + C \implies y = -\log x - 1 + Cx \tag{2.2.5.2}$$

The integrating factor is $\frac{1}{x}$. The following code verifies this with a numerical solution of the ODE in Fig. 2.2.5.1.

codes/cbXIIq14.py

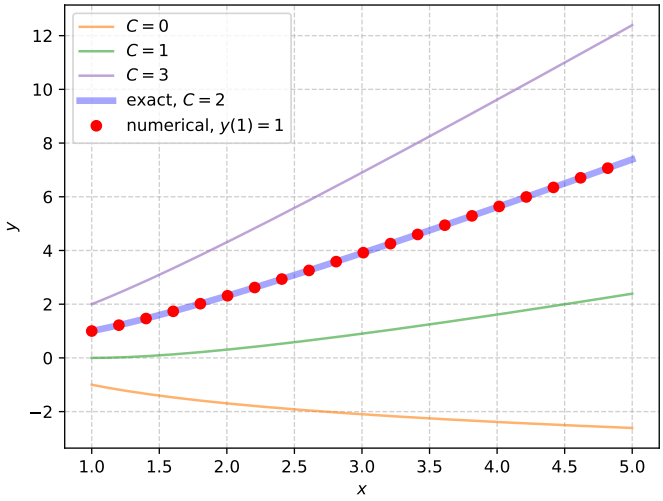


Fig. 2.2.5.1. Numerical solution with $y(1) = 1$ (so $C = 2$) against the closed form

2.2.6 From a pack of 52 cards, 3 cards are drawn at random (without replacement). The probability that they are two red cards and one black card is _____. (CBSE XII 2021)

Solution: Let X be the number of red cards among the three drawn. From (2.1.3.1) with $M = 52$, $K = 26$, $n = 3$,

$$\Pr(X = 2) = \frac{\binom{26}{2}\binom{26}{1}}{\binom{52}{3}} = \frac{325 \times 26}{22100} = \frac{13}{34} \tag{2.2.6.1}$$

The complete PMF is

$$(p_X(0),\ p_X(1),\ p_X(2),\ p_X(3)) = \left(\frac{2}{17}, \frac{13}{34}, \frac{13}{34}, \frac{2}{17}\right) \tag{2.2.6.2}$$

which sums to 1. The following code verifies this in Fig. 2.2.6.1.

codes/cbXIIq15.py

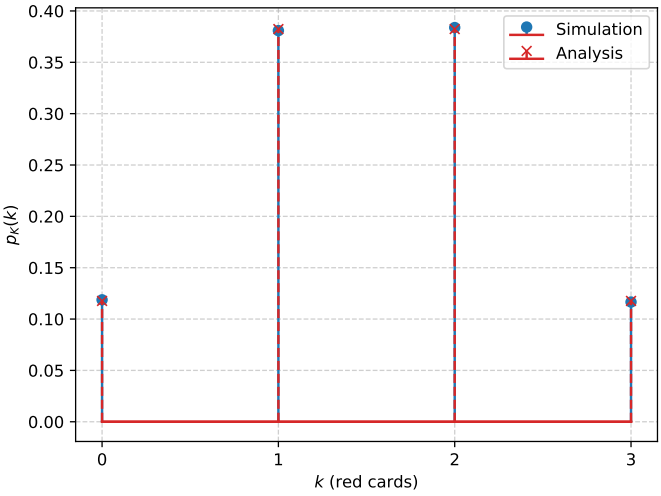


Fig. 2.2.6.1. PMF of the number of red cards in three draws: Simulation vs Analysis

2.2.7 If $\begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}$ is a symmetric matrix, then find the value of x. (CBSE XII 2021)

Solution: Let the matrix be $\mathbf{A}$. From (2.1.10.1), the skew-symmetric part is

$$\mathbf{K} = \frac{\mathbf{A} - \mathbf{A}^\top}{2} = \frac{5-x}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{2.2.7.1}$$

since $(x+2) - (2x-3) = 5 - x$. The matrix is symmetric iff $\mathbf{K} = \mathbf{0}$,

$$5 - x = 0 \implies x = 5, \qquad \mathbf{A} = \begin{pmatrix} 4 & 7 \\ 7 & 6 \end{pmatrix} \tag{2.2.7.2}$$

The following code verifies this in Fig. 2.2.7.1.

codes/cbXIIq19a.py

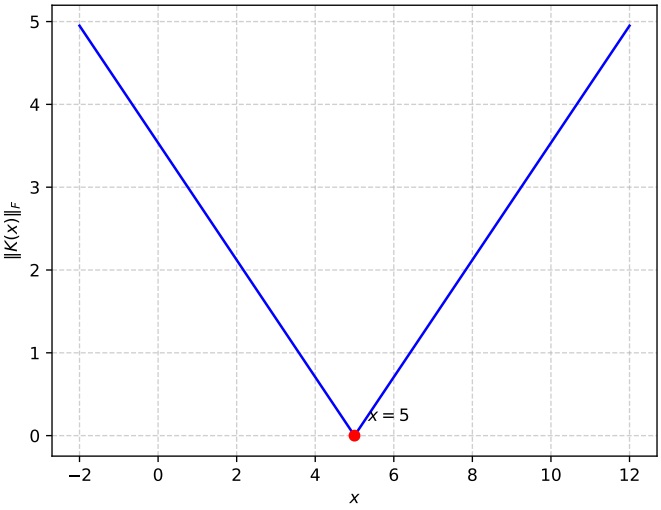


Fig. 2.2.7.1. Norm of the skew-symmetric part $\mathbf{K}(x)$, zero at $x = 5$

2.2.8 If $\mathbf{A}$ is a square matrix such that $\mathbf{A}^2 = \mathbf{A}$, then find $(2\mathbf{I} + \mathbf{A})^3 - 19\mathbf{A}$. (CBSE XII 2021)

Solution: Let $p(\lambda) = (2+\lambda)^3 - 19\lambda = \lambda^3 + 6\lambda^2 - 7\lambda + 8$. Dividing by $\lambda^2 - \lambda$,

$$p(\lambda) = \left(\lambda^2 - \lambda\right)(\lambda + 7) + 8 \tag{2.2.8.1}$$

so $r_1 = 0$ and $r_0 = 8$. Since $\mathbf{A}^2 - \mathbf{A} = \mathbf{0}$, from (2.1.11.1),

$$(2\mathbf{I} + \mathbf{A})^3 - 19\mathbf{A} = \left(\mathbf{A}^2 - \mathbf{A}\right)(\mathbf{A} + 7\mathbf{I}) + 8\mathbf{I} = 8\mathbf{I} \tag{2.2.8.2}$$

This agrees with the eigenvalues of $\mathbf{A}$ being 0 or 1, where $p(0) = p(1) = 8$. The following code verifies this for an orthogonal and an oblique projection matrix in Fig. 2.2.8.1.

codes/cbXIIq19b.py

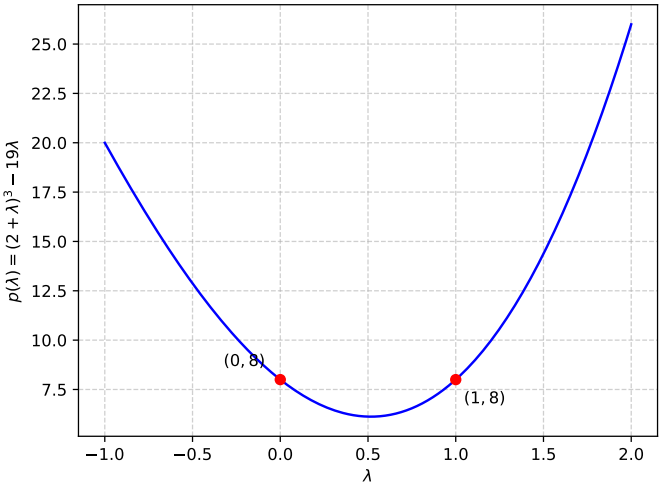


Fig. 2.2.8.1. $p(\lambda) = (2+\lambda)^3 - 19\lambda$ takes the value 8 at $\lambda = 0$ and $\lambda = 1$

2.2.9 Let W denote the set of words in the English dictionary. Define the relation R by $R = \{(x, y) \in W \times W : x \text{ and } y \text{ have at least one letter in common}\}$. Show that R is reflexive and symmetric, but not transitive. (CBSE XII 2021)
Solution: From (2.1.12.1), take $\mathbf{M} = \left(\mathbf{L}\mathbf{L}^\top > 0\right)$, where $L_{ik} = 1$ iff word i contains letter k.

a) Reflexive: $\left(\mathbf{L}\mathbf{L}^\top\right)_{ii} = \sum_k L_{ik}^2$ is the number of distinct letters of word i, which is at least 1 for every word, so $M_{ii} = 1$.

b) Symmetric: $\mathbf{L}\mathbf{L}^\top$ is symmetric, so $\mathbf{M} = \mathbf{M}^\top$.

c) Not transitive: take $x =$ "at", $y =$ "tip", $z =$ "pin". The words "at" and "tip" share t, and "tip" and "pin" share i, p, so $\left(\mathbf{M}^2\right)_{xz} > 0$. But "at" and "pin" have no common letter, so $M_{xz} = 0$ and $\left(\mathbf{M}^2 > 0\right) \not\subseteq (\mathbf{M} > 0)$.

The following code verifies this on a small set of words in Fig. 2.2.9.1.

codes/cbXIIq21a.py

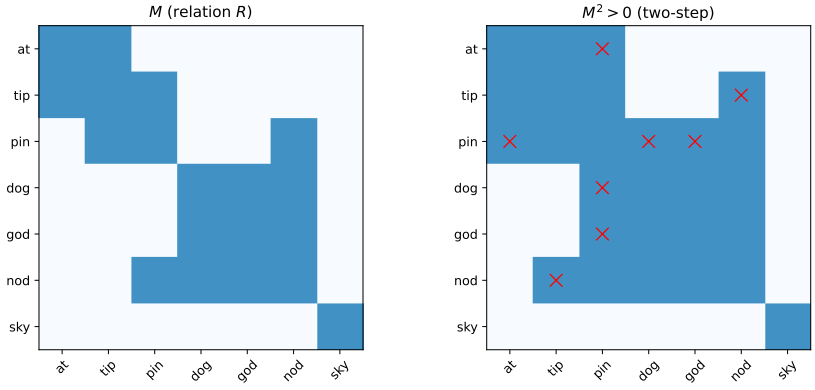


Fig. 2.2.9.1. $\mathbf{M}$ and $\left(\mathbf{M}^2 > 0\right)$ for a few words; crosses mark violations of transitivity

2.2.10 For the matrix $\mathbf{A} = \begin{pmatrix} 2 & 3 \\ -4 & -6 \end{pmatrix}$, verify $\mathbf{A}\,(\operatorname{adj}\mathbf{A}) = (\operatorname{adj}\mathbf{A})\,\mathbf{A} = |\mathbf{A}|\,\mathbf{I}$. (CBSE XII 2021)

Solution: Here $\operatorname{tr}(\mathbf{A}) = 2-6 = -4$ and $|\mathbf{A}| = -12+12 = 0$, so $\mathbf{A}$ is singular and the inverse formula cannot be used. From (2.1.13.1), Cayley-Hamilton gives

$$\mathbf{A}^2 + 4\mathbf{A} = \mathbf{0} \tag{2.2.10.1}$$

(which can be checked directly: $\mathbf{A}^2 = \begin{pmatrix} -8 & -12 \\ 16 & 24 \end{pmatrix} = -4\mathbf{A}$). From (2.1.13.2),

$$\operatorname{adj}\mathbf{A} = -4\mathbf{I} - \mathbf{A} = \begin{pmatrix} -6 & -3 \\ 4 & 2 \end{pmatrix} \tag{2.2.10.2}$$

which is the transposed cofactor matrix. Then

$$\mathbf{A}\,(\operatorname{adj}\mathbf{A}) = -4\mathbf{A} - \mathbf{A}^2 = -\left(\mathbf{A}^2 + 4\mathbf{A}\right) = \mathbf{0} = |\mathbf{A}|\,\mathbf{I} \tag{2.2.10.3}$$

$$(\operatorname{adj}\mathbf{A})\,\mathbf{A} = -4\mathbf{A} - \mathbf{A}^2 = \mathbf{0} = |\mathbf{A}|\,\mathbf{I} \tag{2.2.10.4}$$

Geometrically, the columns of $\operatorname{adj}\mathbf{A}$ are multiples of $(3,\ -2)^\top$, which spans the null space of $\mathbf{A}$, so $\mathbf{A}$ maps them to $\mathbf{0}$. The following code verifies this in Fig. 2.2.10.1.

codes/cbXIIq22.py

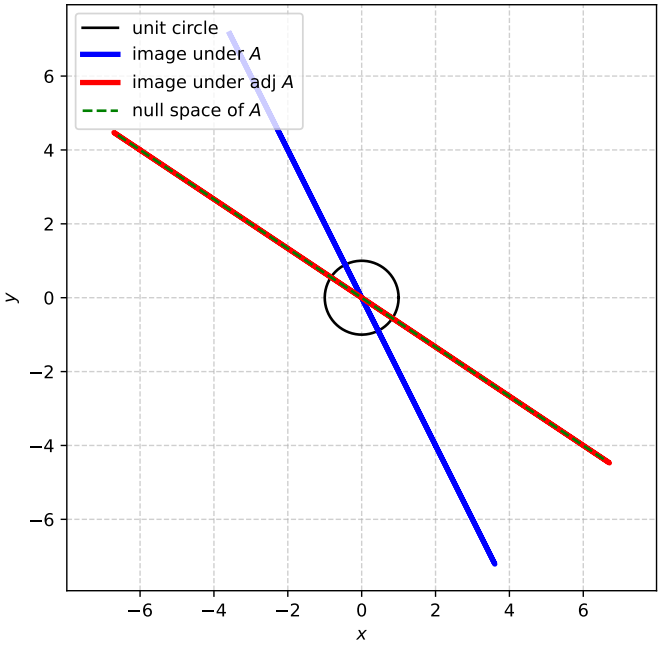


Fig. 2.2.10.1. Images of the unit circle under **A** and adj **A**; the latter lies in the null space of **A**

2.2.11 A bag contains 19 tickets numbered 1 to 19. A ticket is drawn at random and then another ticket is drawn without replacing the first one. Find the probability distribution of the number of even numbers on the tickets. (CBSE XII 2021)
Solution: Let X be the number of even tickets. Among $1,\ldots,19$ there are 9 even and 10 odd tickets, so from (2.1.3.1) with $M=19$, $K=9$, $n=2$,

$$p_X(k)=\frac{\binom{9}{k}\binom{10}{2-k}}{\binom{19}{2}},\quad \binom{19}{2}=171 \tag{2.2.11.1}$$

k	0	1	2
$p_X(k)$	$\frac{45}{171}=\frac{5}{19}$	$\frac{90}{171}=\frac{10}{19}$	$\frac{36}{171}=\frac{4}{19}$

The probabilities sum to 1, and the mean is $E[X]=\frac{2\cdot9}{19}=\frac{18}{19}$. The following code verifies this in Fig. 2.2.11.1.

codes/cbXIIq26a.py

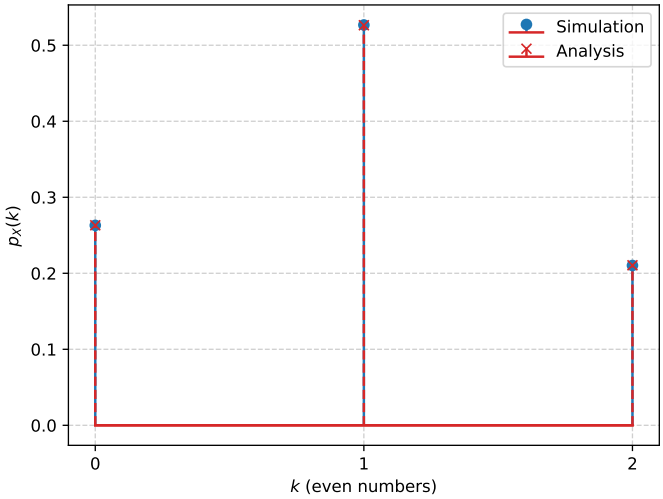


Fig. 2.2.11.1. PMF of the number of even tickets: Simulation vs Analysis

2.2.12 Find the probability distribution of the number of successes in two tosses of a die, where a success is defined as "number greater than 5". (CBSE XII 2021)
Solution: A success is the face 6, so $p = \frac{1}{6}$. Let X be the number of successes in $n = 2$ independent tosses. From (2.1.2.1),

$$p_X(k) = \binom{2}{k}\left(\frac{1}{6}\right)^k\left(\frac{5}{6}\right)^{2-k} \tag{2.2.12.1}$$

k	0	1	2
$p_X(k)$	$\frac{25}{36}$	$\frac{10}{36} = \frac{5}{18}$	$\frac{1}{36}$

This is also the convolution $\left(\frac{5}{6},\ \frac{1}{6}\right) * \left(\frac{5}{6},\ \frac{1}{6}\right)$, with mean $np = \frac{1}{3}$ and variance $np(1-p) = \frac{5}{18}$. The following code verifies this in Fig. 2.2.12.1.

codes/cbXIIq26b.py

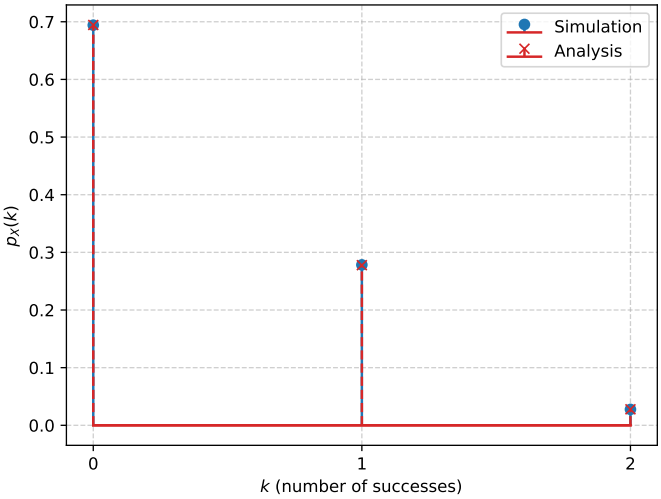


Fig. 2.2.12.1. PMF of the number of sixes in two tosses: Simulation vs Analysis

2.2.13 Find the equation of the line joining $A\,(1,3)$ and $B\,(0,0)$ using determinants. Also find k if $D\,(k,0)$ is a point such that the area of $\triangle ABD$ is 3 square units. (CBSE XII 2021)

Solution: From (2.1.7.1), the line AB is

$$\begin{vmatrix} x & y & 1 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = x\,(3-0) - y\,(1-0) + 1\,(0-0) = 3x - y = 0 \tag{2.2.13.1}$$

The area of $\triangle ABD$ is

$$\Delta = \frac{1}{2}\left|\begin{vmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ k & 0 & 1 \end{vmatrix}\right| = \frac{1}{2}\,|3k| = 3 \implies |k| = 2 \tag{2.2.13.2}$$

$$\therefore k = \pm 2 \tag{2.2.13.3}$$

The following code verifies this in Fig. 2.2.13.1.

codes/cbXIIq28b.py

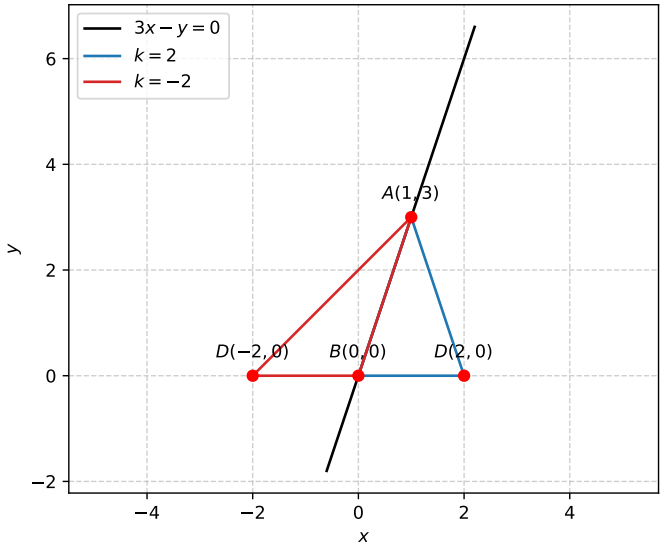


Fig. 2.2.13.1. The line $3x - y = 0$ and the triangles ABD for $k = \pm 2$

2.2.14 Prove that three points A, B and C with position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively are collinear if and only if $(\mathbf{b} \times \mathbf{c}) + (\mathbf{c} \times \mathbf{a}) + (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$. (CBSE XII 2021)
Solution: Let $\mathbf{S} = \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} + \mathbf{a} \times \mathbf{b}$. From (2.1.6.1),

$$\mathbf{S} = (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) \tag{2.2.14.1}$$

This vanishes iff $\mathbf{b} - \mathbf{a}$ and $\mathbf{c} - \mathbf{a}$ are parallel, that is (for $A \neq B$) iff $\mathbf{c} - \mathbf{a} = \kappa\,(\mathbf{b} - \mathbf{a})$ for some κ, which says that C lies on the line AB. If $A = B$ both statements hold trivially. Hence A, B, C are collinear iff $\mathbf{S} = \mathbf{0}$. Moreover $\|\mathbf{S}\| = 2\,\text{area}\,(\triangle ABC)$.
For $\mathbf{a} = (1,\ 0,\ 2)^\top$ and $\mathbf{b} = (3,\ 1,\ 1)^\top$, the point $\mathbf{c} = \mathbf{a} + 2.5\,(\mathbf{b} - \mathbf{a})$ gives $\mathbf{S} = \mathbf{0}$, while $\mathbf{c} = (2,\ 3,\ 4)^\top$ gives $\mathbf{S} = (5,\ -5,\ 5)^\top$ and $\frac{1}{2}\,\|\mathbf{S}\| = 4.33$, which equals the area from Heron's formula. The following code verifies this in Fig. 2.2.14.1.

codes/cbXIIq29.py

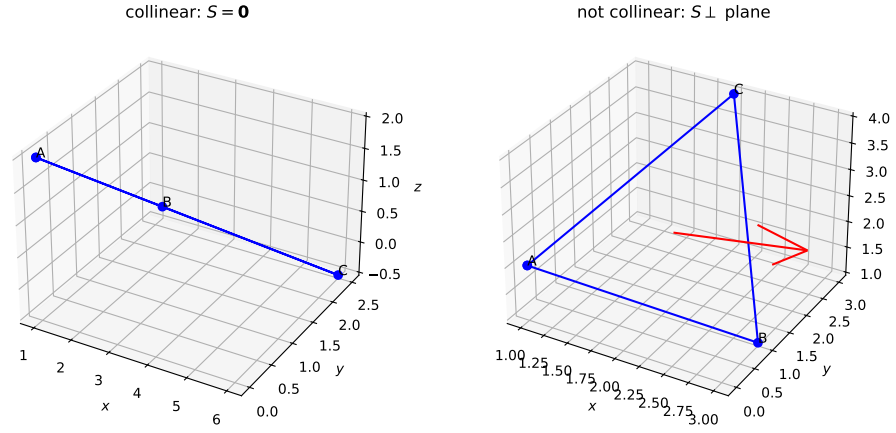


Fig. 2.2.14.1. Collinear points ($\mathbf{S} = \mathbf{0}$) and a triangle with $\mathbf{S}$ normal to its plane

2.2.15 Find the shortest distance between the lines $\frac{x-1}{2} = \frac{y+1}{3} = z$ and $\frac{x+1}{5} = \frac{y-2}{1}$, $z = 2$, and hence write whether the lines are intersecting or not. (CBSE XII 2021)

Solution: From (2.1.5.1), the lines are $\mathbf{x} = \mathbf{a}_1 + s\mathbf{m}_1$ and $\mathbf{x} = \mathbf{a}_2 + t\mathbf{m}_2$ with

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \ \mathbf{m}_1 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}, \ \mathbf{m}_2 = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} \tag{2.2.15.1}$$

$$\mathbf{b} = \mathbf{a}_2 - \mathbf{a}_1 = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix} \tag{2.2.15.2}$$

The normal equations in (2.1.8.1) are

$$\begin{pmatrix} 14 & 13 \\ 13 & 26 \end{pmatrix} \begin{pmatrix} s \\ -t \end{pmatrix} = \begin{pmatrix} 7 \\ -7 \end{pmatrix} \implies s = \frac{7}{5}, \quad t = \frac{63}{65} \tag{2.2.15.3}$$

The closest points are $(3.8,\ 3.2,\ 1.4)^\top$ on the first line and $(3.846,\ 2.969,\ 2)^\top$ on the second. The distance is $\|\mathbf{Mz} - \mathbf{b}\|$, and from (2.1.8.2), with $\mathbf{m}_1 \times \mathbf{m}_2 = (-1,\ 5,\ -13)^\top$ and $\mathbf{b}^\top (\mathbf{m}_1 \times \mathbf{m}_2) = -9$,

$$d = \frac{|-9|}{\sqrt{195}} = \frac{9}{\sqrt{195}} \approx 0.645 \tag{2.2.15.4}$$

Since $d \neq 0$, the lines do not intersect (they are skew). The following code verifies this in Fig. 2.2.15.1.

codes/cbXIIq30a.py

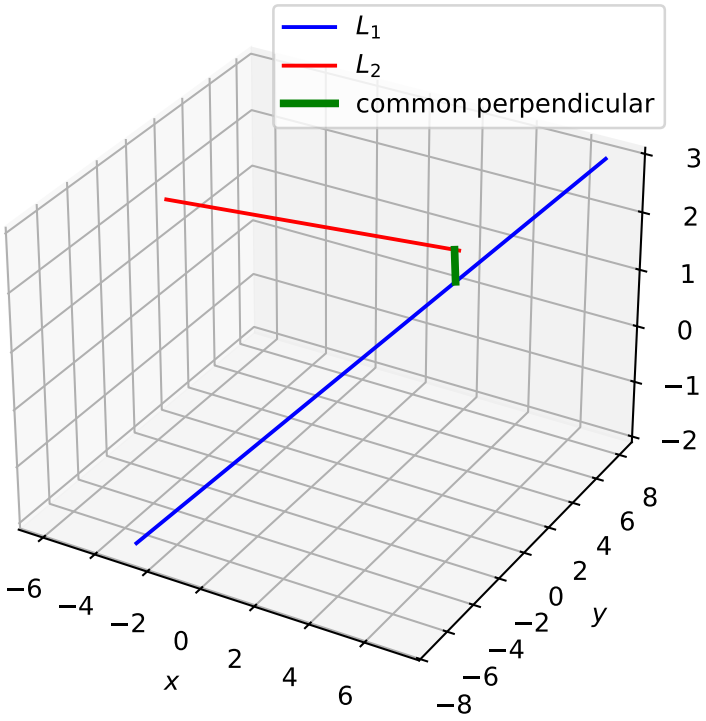


Fig. 2.2.15.1. Skew lines and their common perpendicular

2.2.16 Find the equation of the plane through the line of intersection of the planes $\mathbf{r}\cdot\left(\hat{i}+3\hat{j}\right)+6=0$ and $\mathbf{r}\cdot\left(3\hat{i}-\hat{j}-4\hat{k}\right)=0$, which is at a unit distance from the origin. (CBSE XII 2021)

Solution: The planes are $\mathbf{n}_1^\top\mathbf{x}+e_1=0$ and $\mathbf{n}_2^\top\mathbf{x}+e_2=0$ with

$$\mathbf{n}_1=\begin{pmatrix}1\\3\\0\end{pmatrix},\ e_1=6,\qquad \mathbf{n}_2=\begin{pmatrix}3\\-1\\-4\end{pmatrix},\ e_2=0 \tag{2.2.16.1}$$

From (2.1.9.1), the planes through the line of intersection have $\mathbf{n}=\mathbf{n}_1+\lambda\mathbf{n}_2$ and $e=6$. From (2.1.9.2), the distance from the origin is 1 iff $\|\mathbf{n}\|^2=e^2$,

$$\|\mathbf{n}_1+\lambda\mathbf{n}_2\|^2=\mathbf{n}_1^\top\mathbf{n}_1+2\lambda\mathbf{n}_1^\top\mathbf{n}_2+\lambda^2\mathbf{n}_2^\top\mathbf{n}_2=10+0+26\lambda^2=36\implies\lambda=\pm1 \tag{2.2.16.2}$$

$$\lambda=1:\ 4x+2y-4z+6=0\implies 2x+y-2z+3=0 \tag{2.2.16.3}$$

$$\lambda=-1:\ -2x+4y+4z+6=0\implies x-2y-2z-3=0 \tag{2.2.16.4}$$

Both are at distance $\frac{3}{3}=1$ from the origin. The following code verifies this in

Fig. 2.2.16.1.

codes/cbXIIq30b.py

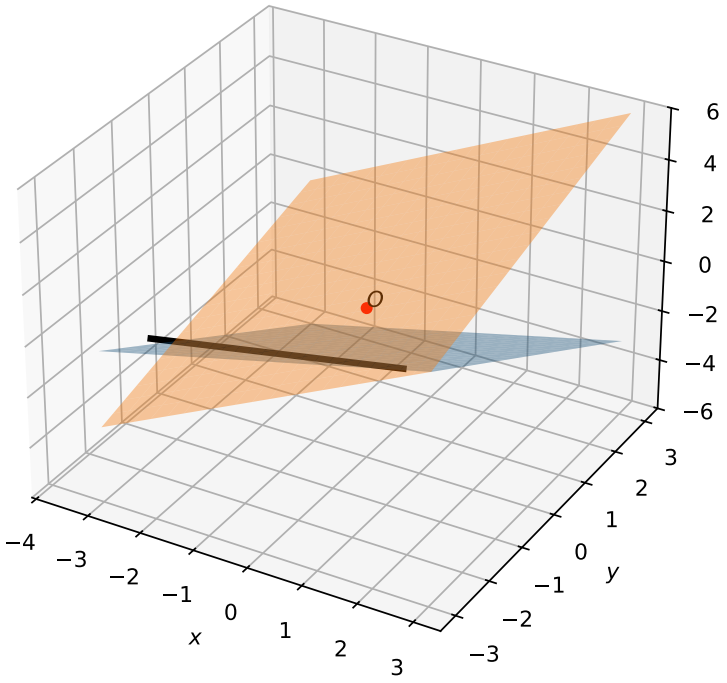


Fig. 2.2.16.1. The two required planes and the line of intersection of the given planes

2.2.17 A bag contains 5 red and 4 black balls, and a second bag contains 3 red and 6 black balls. One of the two bags is selected at random and two balls are drawn at random (without replacement), both of which are found to be red. Find the probability that these two balls are drawn from the second bag. (CBSE XII 2021)

Solution: Let B_i be the event that bag i is chosen, with $\Pr(B_i) = \frac{1}{2}$, and let E be the event that both balls are red. From (2.1.3.1),

$$\Pr(E \mid B_1) = \frac{\binom{5}{2}}{\binom{9}{2}} = \frac{10}{36}, \qquad \Pr(E \mid B_2) = \frac{\binom{3}{2}}{\binom{9}{2}} = \frac{3}{36} \tag{2.2.17.1}$$

From (2.1.4.1),

$$\Pr\left(B_2 \mid E\right) = \frac{\frac{1}{2}\cdot\frac{3}{36}}{\frac{1}{2}\cdot\frac{10}{36}+\frac{1}{2}\cdot\frac{3}{36}} = \frac{3}{13} \tag{2.2.17.2}$$

The following code verifies this in Fig. 2.2.17.1.

codes/cbXIIq32.py

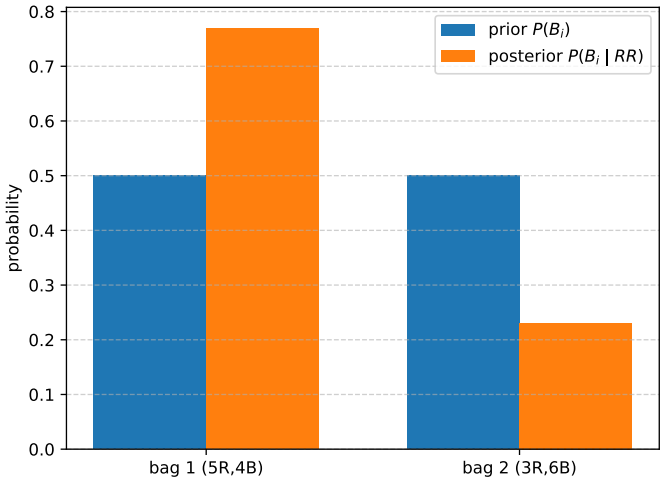


Fig. 2.2.17.1. Prior and posterior probabilities of the two bags

2.2.18 If the area between the curves $x = y^2$ and $x = 4$ is divided into two equal parts by the line $x = a$, then find the value of a using integration. (CBSE XII 2021)
Solution: From (2.1.17.1), the total area is $A\left(4\right) = \frac{4}{3}\cdot 4^{3/2} = \frac{32}{3}$. The line $x = a$ halves it iff

$$A\left(a\right) = \frac{4}{3}a^{3/2} = \frac{1}{2}\cdot\frac{32}{3} \implies a^{3/2} = 4 \tag{2.2.18.1}$$

$$\therefore a = 4^{2/3} = 2^{4/3} \approx 2.52 \tag{2.2.18.2}$$

The following code solves $A\left(a\right) = \frac{1}{2}A\left(4\right)$ numerically and verifies this in Fig. 2.2.18.1.

codes/cbXIIq35a.py

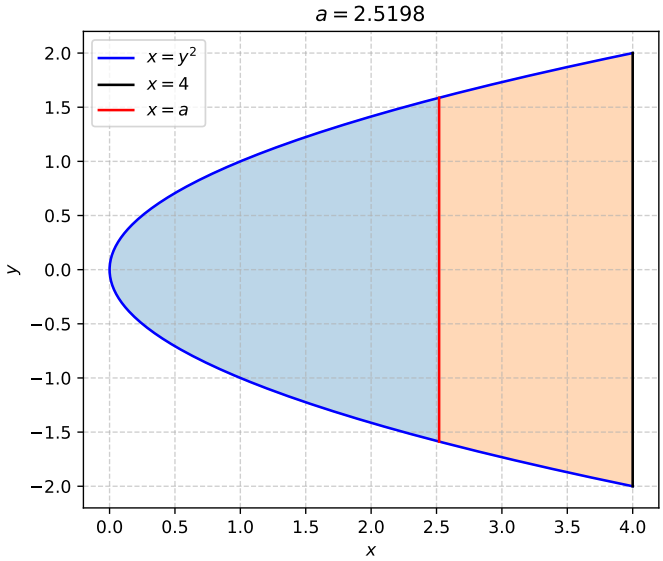


Fig. 2.2.18.1. Region between $x = y^2$ and $x = 4$ divided equally by $x = a$

2.2.19 Find $\int \frac{x}{(x-1)^2(x+2)}dx$. (CBSE XII 2021)

Solution: From (2.1.18.1), $x = A(x-1)(x+2) + B(x+2) + C(x-1)^2$. Matching the coefficients of x^2, x and 1,

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & -2 \\ -2 & 2 & 1 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \implies \begin{pmatrix} A \\ B \\ C \end{pmatrix} = \begin{pmatrix} \frac{2}{9} \\ \frac{1}{3} \\ -\frac{2}{9} \end{pmatrix} \tag{2.2.19.1}$$

Hence

$$\int \frac{x\,dx}{(x-1)^2(x+2)} = \frac{2}{9}\ln|x-1| - \frac{1}{3(x-1)} - \frac{2}{9}\ln|x+2| + c \tag{2.2.19.2}$$

The following code verifies this, including $\int_2^5 = 0.4337$ against the antiderivative, in Fig. 2.2.19.1.

codes/cbXIIq35b.py

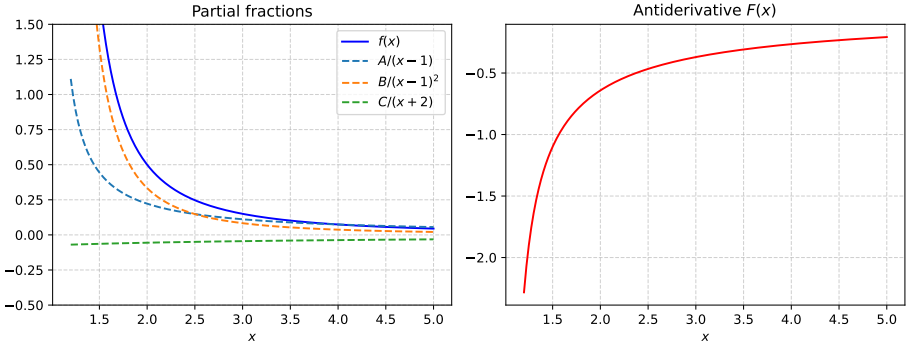


Fig. 2.2.19.1. Partial fractions of the integrand and its antiderivative